Ministry of Education and Science of Ukraine



Zhytomyr Agrotechnical Professional College

Department of Automotive Transport

CALCULATION AND EXPLANATORY REPORT

Vehicle powertrain design calculation

Vehicle powertrain design calculation

ICE Design · 0.8.6 · complete current calculation

Zhytomyr

2026

Contents

1. Introduction and problem statement
2. Design inputs
3. Calculation assumptions and initial dimensions
4. Intake, compression, combustion, expansion, and exhaust
5. Performance, adopted dimensions, and characteristics
6. Kinematics, dynamics, and auxiliary systems
11. EPA maps and driving scenarios
12. Strength and fatigue life: first approximation
13. Simplified control strategy
15. Comparison with measurements
16. Conclusions, sources, and limitations

1 Introduction and problem statement

How to read the notation

Symbols follow full quantity names: power (P), torque (M), rotational speed (n), vehicle speed (v), mass (m), force (F), energy (E), voltage (U), current (I), resistance (R), and efficiency (η). Lowercase p denotes pressure. A subscript identifies the component or operating mode, for example, P_{S1} — continuous power, M_{peak} — peak torque.

An internal combustion engine (ICE) may use spark ignition (SI) or compression ignition (CI). PFI means port fuel injection; DI means direct injection, and GDI means gasoline direct injection. The slider-crank mechanism and valve gear are distinguished; TDC and BDC mean top and bottom dead centre, and CA means crank angle.

BEV means battery electric vehicle; HEV means hybrid electric vehicle; PHEV means plug-in hybrid electric vehicle. P2 is a parallel layout with an electric machine between the engine and gearbox. The battery's SOC is its state of charge; OCV is open-circuit voltage (U_{oc}). Equivalent battery internal resistance is R_{int}; this is an ohmic internal-resistance model. NMC means lithium-ion nickel manganese cobalt oxide; LFP means lithium iron phosphate, $LiFePO_4$. Chemistry alone does not determine pack voltage, specific energy, or current limits.

PMSM means permanent-magnet synchronous motor. d and q are the direct and quadrature axes of a rotating coordinate system; L_{d} and L_{q} — axis inductances; ψ is flux linkage. RMS means root mean square; for a sinusoid, peak amplitude equals √2 times RMS. In phase voltage, peak denotes amplitude; in power/torque characteristics, it denotes short-duration duty. DC means direct current; AC means alternating current. S1 is continuous duty under steady thermal conditions; peak is short-duration peak duty. SVPWM means space-vector pulse-width modulation.

WLTP means Worldwide harmonized Light vehicles Test Procedure; its driving cycle is WLTC, Worldwide harmonized Light vehicles Test Cycle. PER means pure electric range. UBE means usable battery energy, measured here at the terminals; energy consumption per distance is ($\mathrm{EC}_{\mathrm{DC}}$). CSS means constant-speed segment; M and E denote intermediate and final segments. The index in WLTC_1, WLTC_2 denotes the repetition number, not another cycle class.

EPA is the US Environmental Protection Agency. ALPHA means Advanced Light-duty Powertrain and Hybrid Analysis. UDDS is the Urban Dynamometer Driving Schedule; HWFET is the Highway Fuel Economy Test; US06 is the aggressive supplementary schedule. These names identify specific speed profiles.

ECU means electronic control unit; BMS means battery management system; OBD2 means second-generation on-board diagnostics. RMSE means root mean square error. The International System of Units (SI) is distinguished by context from SI meaning spark ignition. File formats are PDF (Portable Document Format), CSV (Comma-Separated Values), JSON (JavaScript Object Notation), SVG (Scalable Vector Graphics), HTML (HyperText Markup Language); ZIP is an archive format.

This is a general design and verification calculation for an engine and powertrain. First specify the required performance and adopt the parameters, then check the processes, dimensions, and subsequent calculations.

Selecting a petrol engine, diesel engine, hybrid, or BEV changes the active fields and chapters. The carburettor branch is excluded. Modern injection and boosting are represented by the available model; Euro 5–6 emissions compliance is not numerically verified.

P_{req} is a requirement; P_{calc} is a result. Calculated and adopted geometry are stored separately. Changing the inputs makes all dependent results stale until recalculation.

2 Design input data

Course project assignment for Automotive Engines. The principal inputs follow the assignment form; hybrid and electric powertrains additionally specify the system architecture and component requirements.

Supporting assumptions appear beside their respective calculations in subsequent chapters. The initial dataset is filled in; all applicable parameters can be edited.

> Profile: M1 (passenger car) · naturally aspirated petrol. Guideline interval 4500–7000 min⁻¹; current value 5500 min⁻¹. This is not an automatic design limit. [2]

Table 2.1. Inputs.

Field	Adopted	Provenance
I-1 · Powertrain architecture	Internal combustion engine	Given / adopted
I-2 · Ignition method	Spark ignition (petrol engine)	Given / adopted
I-3 · Vehicle category and initial engine profile	M1 (passenger car) · naturally aspirated petrol	Given / adopted [2]
I-6 · Required output-shaft power	110 kW	Given / adopted [2]
I-7 · Speed at which power I-6 is required	5 500 min⁻¹	Given / adopted [2]
I-8 · Geometric compression ratio	10,5	Given / adopted [2]
I-9 · Cylinder arrangement	In-line	Given / adopted [1]
I-10 · Combustion chamber type	Compact undivided	Given / adopted [2]
I-11 · Mixture preparation	Port fuel injection (PFI)	Given / adopted [2]
I-12 · Absolute pressure before the inlet valves	101,3 kPa	Given / adopted [2]
I-13 · Compressor type	No compressor	Given / adopted [2]
I-14 · Cooling system	Liquid	Given / adopted [2]
I-15 · Vehicle type	Passenger car	Given / adopted [2]
I-16 · Mechanism / system design task	Slider-crank mechanism	Given / adopted

3 Calculation assumptions and initial dimensions

The elemental C/H/O composition and oxygen mass fraction in air of 0.23 determine theoretical air demand. The effective heat capacities c_{v}(T)=675+0.25T and R=287 are not chemical-equilibrium properties of the products.

In design mode, the initial p_{e0} determines only the first approximation to D. The iterative solver then finds D at the specified S/D from the brake power of the reintegrated cycle. In verification mode, D and S are specified.

Calculated bore and stroke are not nominal drawing dimensions. For preliminary adoption, D and S may be rounded upwards using increment I-177, normally 1 mm, and recalculated. Production engines need not use integer millimetres: BMW specifies D = 82 mm, S = 94.6 mm [27].

Occupant and payload guidelines

The category describes purpose and mass limits; you specify trip payload. Initial project values and approximate nominal payload from the Automobile adaptation are given below.

Table 3.1. Initial payloads and reference payload by category.

Category	Initial cargo, kg	Reference payload, kg
M1 · Passenger car	200	≈ 546
M2 · Bus to 5 t	500	≈ 800
M3 · Bus above 5 t	1 500	≈ 5 000
N1 · Goods vehicle to 3,5 t	500	≈ 500
N2 · Goods vehicle above 3,5–12 t	2 000	≈ 4 500
N3 · Goods vehicle above 12 t	5 000	≈ 14 000

M1 has up to 8 passenger seats excluding the driver; M2/M3 have more than 8. Tabulated payloads are editable design guidelines, not regulatory limits for every category. Use the actual vehicle's specified payload. [19] [9]

Count occupants and luggage once: if the driver is already included in the adopted kerb mass, do not add the driver to payload again. Changing category must not replace manual I-174.

Table 3.2. Inputs.

Field	Adopted	Provenance
I-19 · Calculation mode	Design — determine dimensions and parameters	Given / adopted
I-20 · Engine cycle	Four-stroke · 720°	Given / adopted
I-21 · Number of cylinders	4	Given / adopted [2]
I-22 · Stroke-to-bore ratio	0,95	Given / adopted [2]
I-23 · Initial brake mean effective pressure estimate	1 000 kPa	Given / adopted [2]
I-26 · Dimension adoption after calculation	Use calculated D and S	Given / adopted
I-27 · Crank-radius-to-rod-length ratio	0,28	Given / adopted [2]
I-28 · Design maximum-pressure limit	14 MPa	Given / adopted [2]
I-29 · Mean piston-speed limit	25 m/s	Given / adopted

Angular speed of the specified output shaft.

Angular speed of the specified output shaft

where n_{req} — Speed at which power I-6 is required, s⁻¹

$$\omega_{\text{req}} = 2 \cdot \pi \cdot n_{\text{req}} \tag{3.1}$$

Substitute the current values in SI units:

$\omega_{\text{req}} = 2 \cdot 3{,}142 \cdot 91{,}67 = 576 \ \text{rad/s}$

The result passes to dependent calculations without being entered again.

The specified power and rotational speed determine torque at the rated point. This does not search the full characteristic for maximum torque.

Required torque at the specified point, not maximum torque

where P_{req} — Required output-shaft power, W; ω_{req} — Angular speed of the specified output shaft, rad/s

$$M_{\text{req}} = \frac{P_{\text{req}}}{\omega_{\text{req}}} \tag{3.2}$$

Substitute the current values in SI units:

$M_{\text{req}} = \frac{110000}{576} = 191 \ \text{N·m}$

The result passes to dependent calculations without being entered again.

The elemental mass fractions determine the theoretical air requirement. The oxygen fraction in air, 0.23, is an assumption of this method. [3]

Theoretical air requirement by mass

where C — Carbon mass fraction in the model fuel; H — Hydrogen mass fraction; O — Oxygen mass fraction

$$l_0 = \frac{\left(\left(\frac{8}{3} \right) \cdot C + 8 \cdot H - O \right)}{0{,}23} \tag{3.3}$$

Substitute the current values in SI units:

$l_0 = \frac{\left(\left(\frac{8}{3} \right) \cdot 0{,}855 + 8 \cdot 0{,}145 - 0 \right)}{0{,}23} = 14{,}96$

The result passes to dependent calculations without being entered again.

Calculated bore: D_{calc} = 91,86 mm.

Calculated stroke: S_{calc} = 87,27 mm.

Initial displacement approximation: V_{seed} = 2,4 L.

Bore adopted for subsequent calculations: 91.86 mm. The adopted value is referenced directly from the preceding result.

Stroke adopted for subsequent calculations: 87.27 mm. The adopted value is referenced directly from the preceding result.

Piston area.

Piston area

where D — Bore adopted for subsequent calculations, m

$$A_{\mathrm{p}} = \frac{\pi \cdot D^{2}}{4} \tag{3.4}$$

Substitute the current values in SI units:

$$A_{\mathrm{p}} = \frac{3{,}142 \cdot 0{,}09186^{2}}{4} = 0{,}006628 \ \mathrm{m^{2}}$$

The result passes to dependent calculations without being entered again.

Swept volume of one cylinder.

Swept volume of one cylinder

where A_{p} — Piston area, m²; S — Stroke adopted for subsequent calculations, m

$$V_{\mathrm{h}} = A_{\mathrm{p}} \cdot S \tag{3.5}$$

Substitute the current values in SI units:

$$V_{\mathrm{h}} = 0{,}006628 \cdot 0{,}08727 = 5{,}784 \times 10^{-4} \ \mathrm{m^{3}}$$

Result in litres: V_{h} = 0,5784 L.

The result passes to dependent calculations without being entered again.

Total engine displacement.

Total engine displacement

where V_{h} — Swept volume of one cylinder, m³; z — Number of cylinders, 1

$$V_{\Sigma} = V_{\mathrm{h}} \cdot z \tag{3.6}$$

Substitute the current values in SI units:

$$V_{\Sigma} = 5{,}784 \times 10^{-4} \cdot 4 = 0{,}002314 \ \mathrm{m^3}$$

Result in litres: V_{Σ} = 2,314 L.

The result passes to dependent calculations without being entered again.

The adopted compression ratio and swept volume determine the geometric clearance volume.

Clearance volume of one cylinder

where V_{h} — Swept volume of one cylinder, m³; ε — Geometric compression ratio

$$V_{\mathrm{c}} = \frac{V_{\mathrm{h}}}{(\,\varepsilon - 1\,)} \tag{3.7}$$

Substitute the current values in SI units:

$$V_{\mathrm{c}} = \frac{5{,}784 \times 10^{-4}}{(\,10{,}5 - 1\,)} = 6{,}089 \times 10^{-5} \ \mathrm{m^3}$$

Result in litres: V_{c} = 0,06089 L.

The result passes to dependent calculations without being entered again.

Connecting rod centre distance.

Connecting rod centre distance

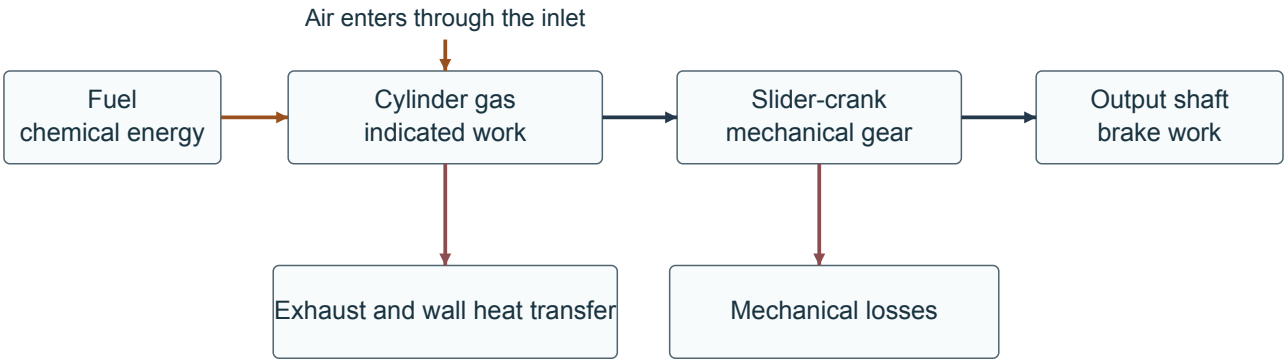
where S — Stroke adopted for subsequent calculations, m; λ — Crank-radius-to-rod-length ratio

$$l = \frac{S}{(\,2 \cdot \lambda\,)} \tag{3.8}$$

Substitute the current values in SI units:

$$l = \frac{0{,}08727}{(\,2 \cdot 0{,}28\,)} = 0{,}1558 \ \mathrm{m}$$

The result passes to dependent calculations without being entered again.



Dark: mechanical; turquoise: electrical; brown: fuel/intake; burgundy: losses.

Figure 3.1. Engine energy conversion and output-shaft boundary. Useful shaft power is below indicated power because of mechanical losses. Wall heat and exhaust energy do not become shaft work. Arrows indicate paths, not energy fractions.

> **M1 (passenger car) · naturally aspirated petrol**
>
> Passenger car: no more than 8 seats excluding the driver; no standing places. [19]
>
> The category determines vehicle purpose, seating, and mass; engine type and speeds are separate design assumptions. M1 and passenger car denote the same category here. Initially: n = 5500 min⁻¹; series 800–6500 min⁻¹.

Initial parameters by engine category

Speeds adopted from the category table [2]. Cylinder count, compression ratio, and boundary conditions are editable design assumptions, not properties determined by vehicle category alone.

Table 3.3. Initial parameters by engine category.

Category / type	n_N, min⁻¹	z	ε	S/D	p_{in}, kPa
M1 (passenger car) · naturally aspirated petrol [2]	5500	4	10,5	0,95	101,3
M1 (passenger car) · boosted petrol	5500	4	10,5	0,95	160
M1 (passenger car) · petrol hybrid profile	5200	4	10,5	0,95	101,3
M2 (bus up to 5 t) · naturally aspirated petrol [2]	5000	4	10,5	0,95	101,3
M3 (bus over 5 t) · naturally aspirated petrol [2]	4000	6	10,5	0,95	101,3
N1 (goods vehicle up to 3.5 t) · naturally aspirated petrol [2]	5000	4	10,5	0,95	101,3
N2 (goods vehicle 3.5–12 t) · naturally aspirated petrol [2]	4200	6	10,5	0,95	101,3
N3 (goods vehicle over 12 t) · naturally aspirated petrol [2]	4000	6	10,5	0,95	101,3

Method guidelines for geometry and piston speed

Historical guidelines from the method [13]; carburettor ranges are not universal recommendations for modern petrol engines.

Table 3.4. Calculation results.

Engine type	Quantity	Range
Carburettor engine	$\frac{S}{D}$	0,7–1,05 [13]
Diesel	$\frac{S}{D}$	0,75–1,15 [13]
Automotive or tractor engine	$\frac{R}{L}$	0,25–0,3 [13]
Carburettor engine passenger car vehicle	c_{m}	12–16 m/s [13]
Carburettor engine truck vehicle	c_{m}	9–13 m/s [13]
Automotive gas engine	c_{m}	7–12 m/s [13]
Automotive diesel	c_{m}	7–14 m/s [13]

Specific heavy-diesel example: 6 cylinders, D = 125 mm, S = 136 mm, ε = 16.3; 168–246 kW at 2100 min^{-1} [18]. This is one design example, not a mandatory configuration for the complete category.

Calculated geometry: D = 91.863 mm, S = 87.27 mm. The calculated geometry was selected for subsequent chapters: D = 91.863 mm, S = 87.27 mm. The cycle and power were recalculated using it.

Table 3.5. Cylinder-bore search iterations.

Iteration	D, mm	Calculated power, kW	Deviation from requirement, %
1	92,992	113,7	3,382
2	91,967	110,3	0,3082
3	91,873	110	0,0277
4	91,864	110	0,002483
5	91,863	110	$2{,}225 \times 10^{-4}$

4 Intake, compression, combustion, expansion, and exhaust

A single-zone 720° cycle is integrated until a periodic state is reached. Valve flows use a compressible-nozzle model; the geometry of the centred slider-crank mechanism is exact. Two inlet and two exhaust valves use a $\sin^2$ lift law without overlap.

PFI: the fresh mixture already contains fuel. GDI/diesel: air enters through the inlet, while fuel mass and its injection energy are added separately. Diesel uses an assumed ignition delay and two Wiebe phases; combustion cannot precede fuel injection.

There is no spray, NOx/soot chemistry, knock, or multizone stratification model. Boosted intake/exhaust pressures are prescribed boundary conditions, not turbocharger selection results.

The single-zone (0D) model represents gas in each cylinder as one homogeneous control volume, with one pressure and mean temperature at each instant. A single zone does not mean a single cylinder. This simplification permits rapid integration of mass and energy balances to determine p–V, work, and power; it does not resolve a flame front, spray, local temperatures, or separate fresh/burnt gas zones. The homogeneous-control-volume principle is described in [21]. This application uses its own simplified heat-capacity and heat-release laws. The dashed line on indicator diagrams represents atmospheric pressure p_0 = 0.101325 MPa.

Table 4.1. Inputs.

Field	Adopted	Provenance
I-30 · Charge temperature before intake	298 K	Given / adopted [2]
I-31 · Absolute exhaust-reservoir pressure	108 kPa	Given / adopted [2]
I-32 · Exhaust backflow temperature	850 K	Given / adopted
I-33 · Effective wall temperature	430 K	Given / adopted
I-34 · Absolute crankcase pressure	100 kPa	Given / adopted
I-35 · Carbon mass fraction in the model fuel	0,855	Given / adopted [3]
I-36 · Hydrogen mass fraction	0,145	Given / adopted [3]
I-37 · Oxygen mass fraction	0	Given / adopted [3]
I-38 · Lower heating value	44 MJ/kg	Given / adopted [3]
I-39 · Fuel density	745 kg/m³	Given / adopted
I-40 · Excess-air ratio	1	Given / adopted [2]
I-41 · Utilised fraction of fuel heat	0,98	Given / adopted
I-52 · Inlet closure	220 ° CA	Given / adopted
I-53 · Exhaust opening	500 ° CA	Given / adopted
I-54 · Inlet-valve diameter / D	0,3975	Given / adopted
I-55 · Exhaust-valve diameter / D	0,3478	Given / adopted
I-56 · Maximum valve lift / D	0,105	Given / adopted
I-57 · Valve discharge coefficient	0,7	Given / adopted
I-58 · Petrol-engine heat-release start	338 ° CA	Given / adopted
I-59 · Petrol-engine heat-release duration	55 ° CA	Given / adopted
I-60 · Wiebe shape exponent	2	Given / adopted
I-71 · Heat-transfer coefficient multiplier	1	Given / adopted
I-72 · Integrator crank-angle step	0,5 °	Given / adopted

At each crank-angle step, the change in working-fluid mass accounts for gas exchange and direct injection.

m is cylinder mass, kg; θ is angle in degrees; n is rotational speed in min⁻¹; ṁ is mass flow in kg/s. The factor 1/(6n) converts seconds to one degree. For PFI, fuel mass is already included in the fresh mixture, so the separate term is zero.

$$\frac{dm}{d\theta} = \frac{\dot{m}_{in} + \dot{m}_{ex}}{6n} + \frac{dm_f}{d\theta} \tag{4.1}$$

Substitution of integrated quantities at a 0.5° step:

$$\Delta m_{res} = -1{,}627 \times 10^{-11} - \left(6{,}092 \times 10^{-4} + -6{,}092 \times 10^{-4} + 0 \right)$$

Residual of the integrated cycle mass balance: ≈ 0 kg.

The first law of thermodynamics is integrated. Heat transfer and mechanical work reduce internal energy; enthalpy flows retain the sign of gas exchange.

U is internal energy, J; Qc is combustion heat, J; Qw is heat transferred to the walls, J; p is absolute pressure, Pa; V is volume, m³; Ḣ is enthalpy flow, W; hf,eff is the effective enthalpy of injected fuel, J/kg. The laws and boundary conditions are specified in the fields of this chapter.

$$\frac{dU}{d\theta} = \frac{dQ_c}{d\theta} - \frac{dQ_w}{d\theta} - p \cdot \frac{dV}{d\theta} + \frac{H_{in} - H_{out}}{6n} + h_{f,eff} \cdot \frac{dm_f}{d\theta} \tag{4.2}$$

Substitution of integrated quantities at a 0.5° step:

$$\Delta U_{res} = -2{,}92 \times 10^{-6} - \left(1659 - 222{,}9 - 694{,}9 + 184{,}4 - 925{,}4 + 0 \right)$$

First-law residual for one cylinder cycle: ≈ 0 J.

The effective-gas equation determines pressure for every calculated state. The maximum below is the maximum in the complete cycle array, not a separately substituted specification value.

R = 287 J/(kg·K) is the assumed effective gas constant; T is the single-zone mean temperature, K. This is not a chemical-equilibrium model of combustion products.

$$p\left(\theta\right) = \frac{m\left(\theta\right) \cdot R \cdot T\left(\theta\right)}{V\left(\theta\right)} \tag{4.3}$$

Substitution of integrated quantities at a 0.5° step:

$$p_{max} = \frac{6{,}448 \times 10^{-4} \cdot 287 \cdot 2524}{6{,}767 \times 10^{-5}}$$

Maximum pressure in the calculated cycle: 6,902 MPa.

The complete indicator-diagram loop gives the indicated work of one cylinder; the low-pressure gas-exchange loop retains its own sign.

W_i is work per four-stroke cycle, J. Numerical tests compare p–V quadrature against an independent torque integral.

$$W_i = \oint pdV = \int_0^{720} p\left(\theta\right) \cdot \frac{dV}{d\theta} d\theta \tag{4.4}$$

Substitution of integrated quantities at a 0.5° step:

$$W_i \approx \sum_k p_k \cdot \Delta V_k = 694{,}9 \text{ J}$$

Indicated work per cylinder over 720°: 694,9 J.

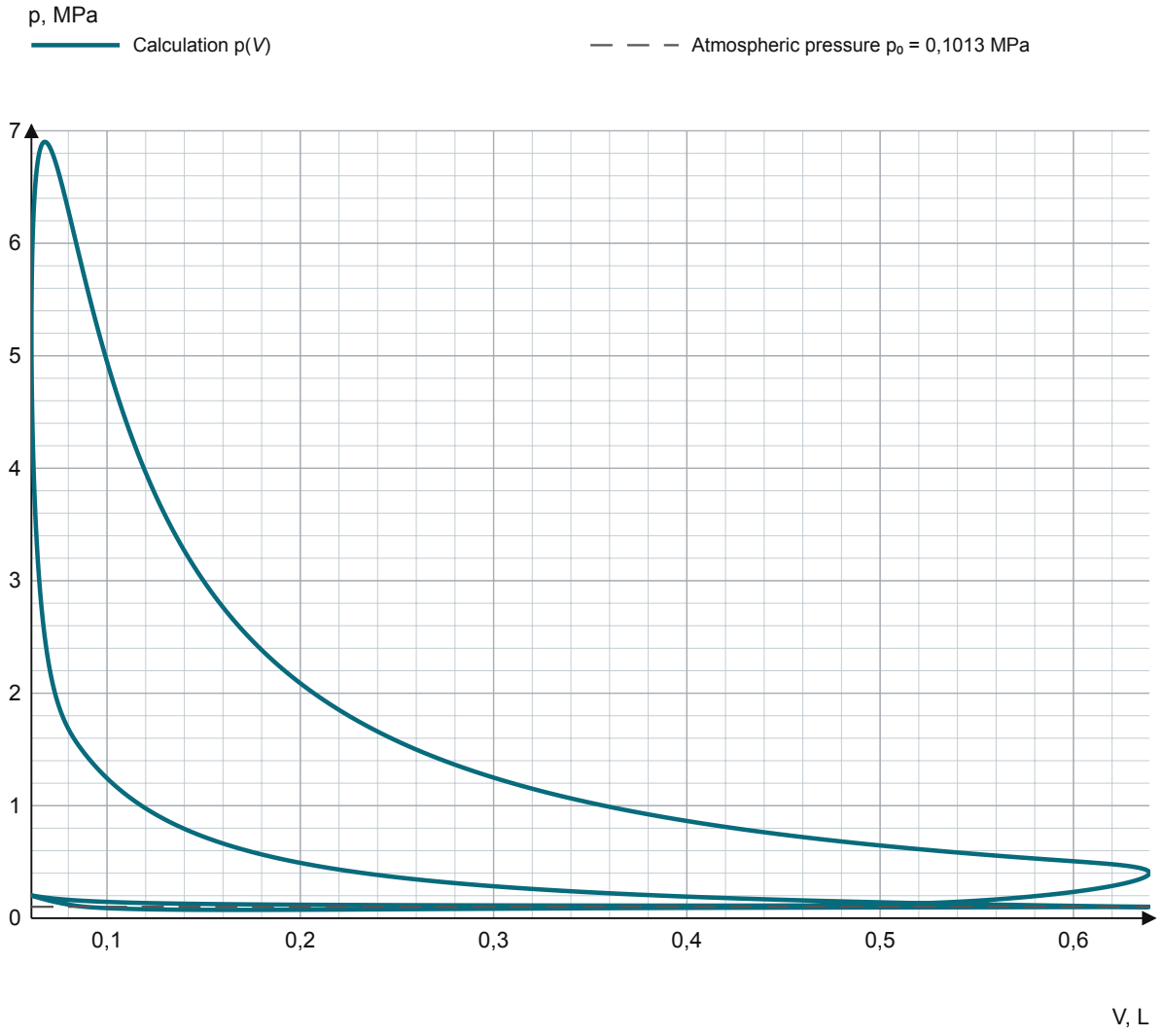


Figure 4.1. Indicator diagram p–V.

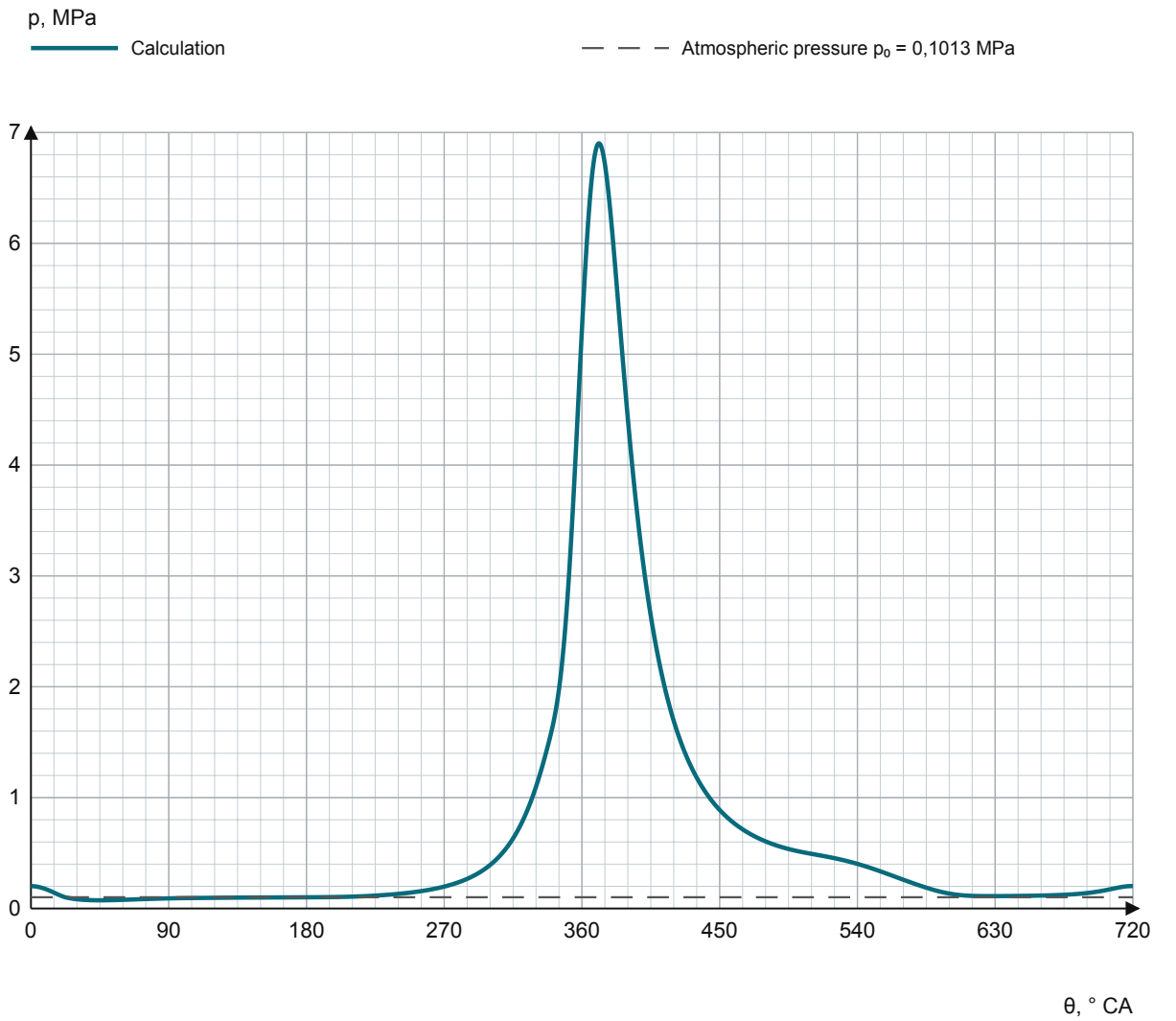


Figure 4.2. Unfolded indicator diagram p–θ, 0–720° CA.

Indicated work per cylinder over 720°: W_{i} = 694,9 J.

Fuel mass per single-cylinder cycle: m_{fc} = 3,847 × 10^{-5} kg.

Friction mean effective pressure: p_{f} = 0,1641 MPa.

Maximum pressure: $p_{\max}$ = 6,902 MPa.

Maximum mean temperature: $T_{\max}$ = 2 725 K.

Heat flow to the walls: Q_{w} = 40,87 kW.

Net enthalpy flow through gas exchange and fuel injection: H_{net} = 135,8 kW.

Energy not released as heat: Q_{unburned} = 6,216 kW.

Pressure at inlet closure: p_{a} = 0,1125 MPa.

Temperature at inlet closure: T_{a} = 357,6 K.

Pressure at heat-release start: p_{c} = 1,481 MPa.

Temperature at heat-release start: T_{c} = 701,6 K.

Pressure at exhaust opening: p_b = 0,5199 MPa.

Temperature at exhaust opening: T_b = 1 653 K.

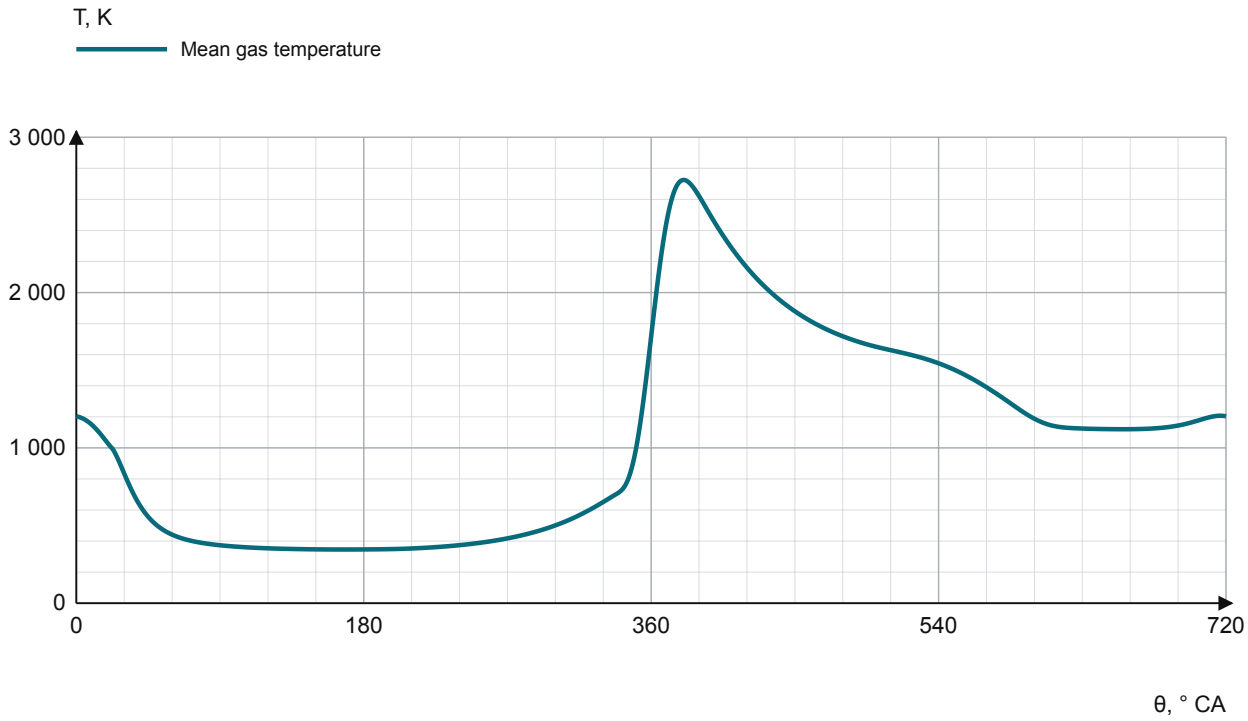


Figure 4.3. Mean working-fluid temperature.

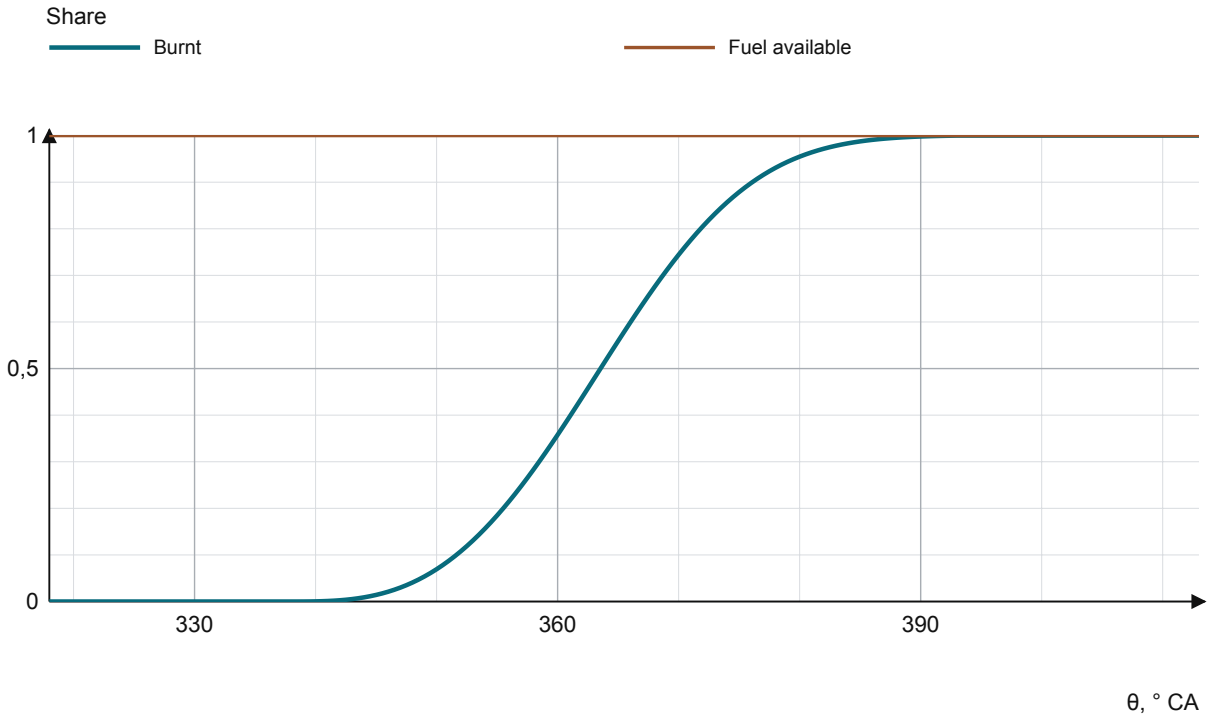


Figure 4.4. Fuel injection and heat-release fraction.

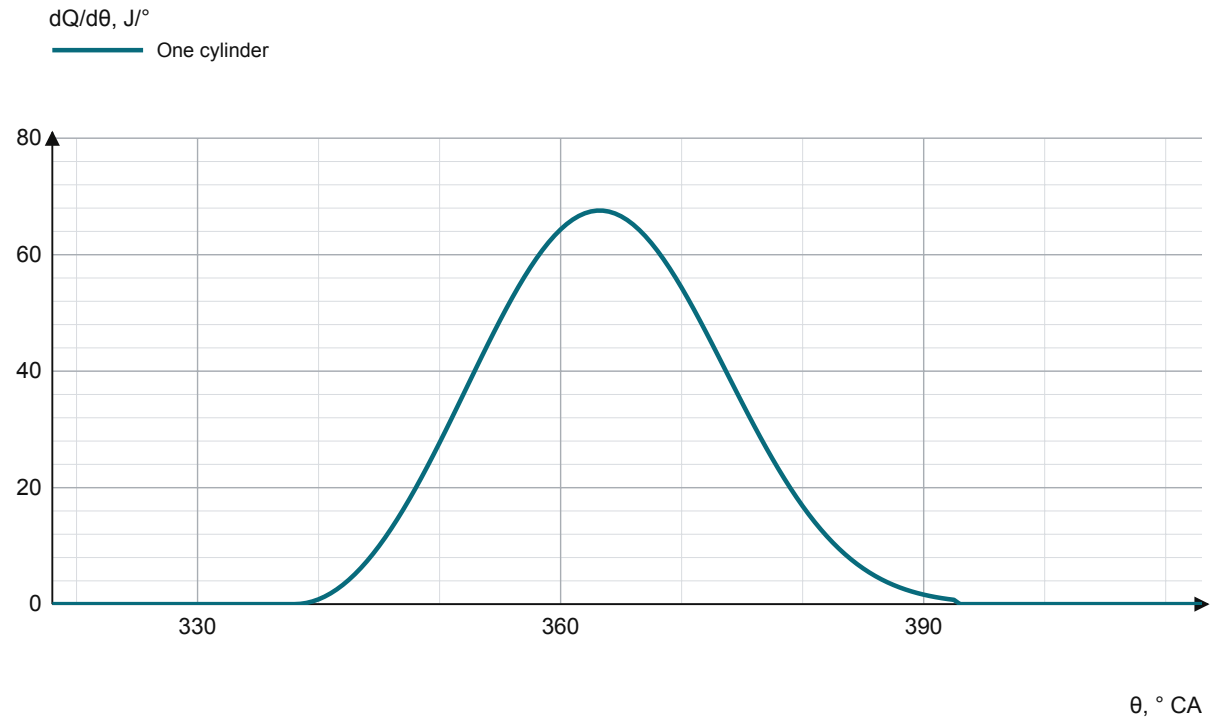


Figure 4.5. Heat release versus crank angle.

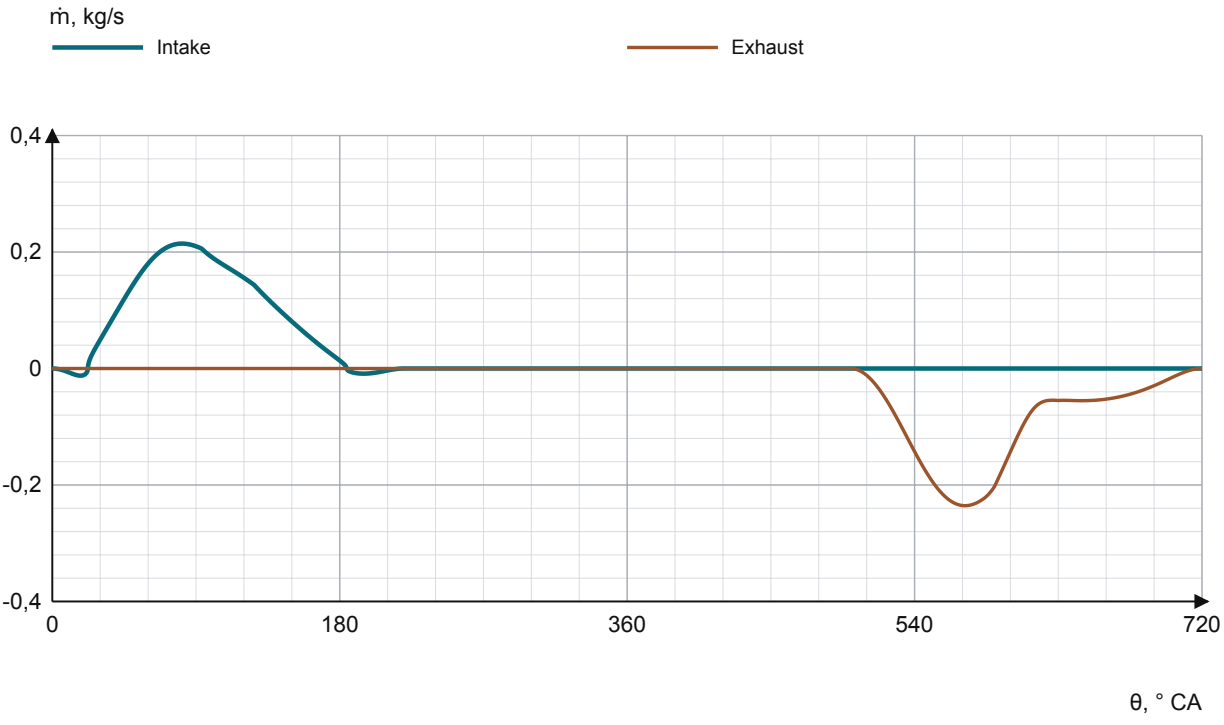


Figure 4.6. Gas exchange through the valves.

5 Performance, adopted dimensions, and characteristics

p–V work determines indicated power; mechanical losses are deducted separately. Fuel consumption follows from the cylinder charge and fuel delivery, rather than a brand-specific coefficient.

The speed series is reintegrated at every rotational speed using the same adopted control laws. It is not an optimised manufacturer's full-load characteristic. Load points are explicitly modelled; a missing EPA-map value is not replaced by zero consumption.

The separate Powertrain module accepts numerical packages with component identity, operating envelopes, and data domains. US Environmental Protection Agency (EPA) packages are installed in Data/EPA; each package retains its own provenance and limitations. The characteristic calculated by this Design module remains modelled, not measured.

Table 5.1. Inputs.

Field	Adopted	Provenance
I-73 · Constant mechanical-loss term	40 kPa	Given / adopted
I-74 · Maximum-pressure coefficient	0,005	Given / adopted
I-75 · Mean piston-speed coefficient	4 000 Pa·s/m	Given / adopted
I-76 · Squared piston-speed coefficient	100 Pa·s²/m²	Given / adopted
I-77 · Lower speed-series boundary	800 min⁻¹	Given / adopted [2]
I-78 · Upper speed-series boundary	6 500 min⁻¹	Given / adopted [2]

Brake power, torque, and efficiency

Speed characteristics and load points are calculated for the current engine. The efficiency map follows from shaft power and fuel consumption at these points; missing points are not invented.

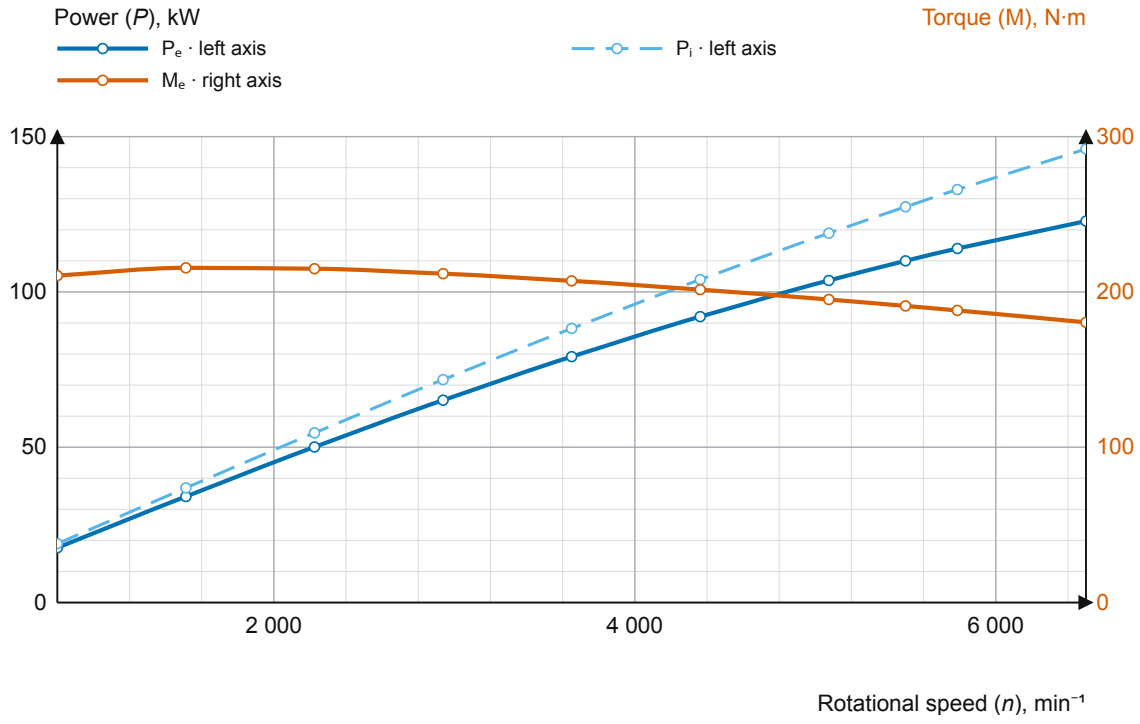
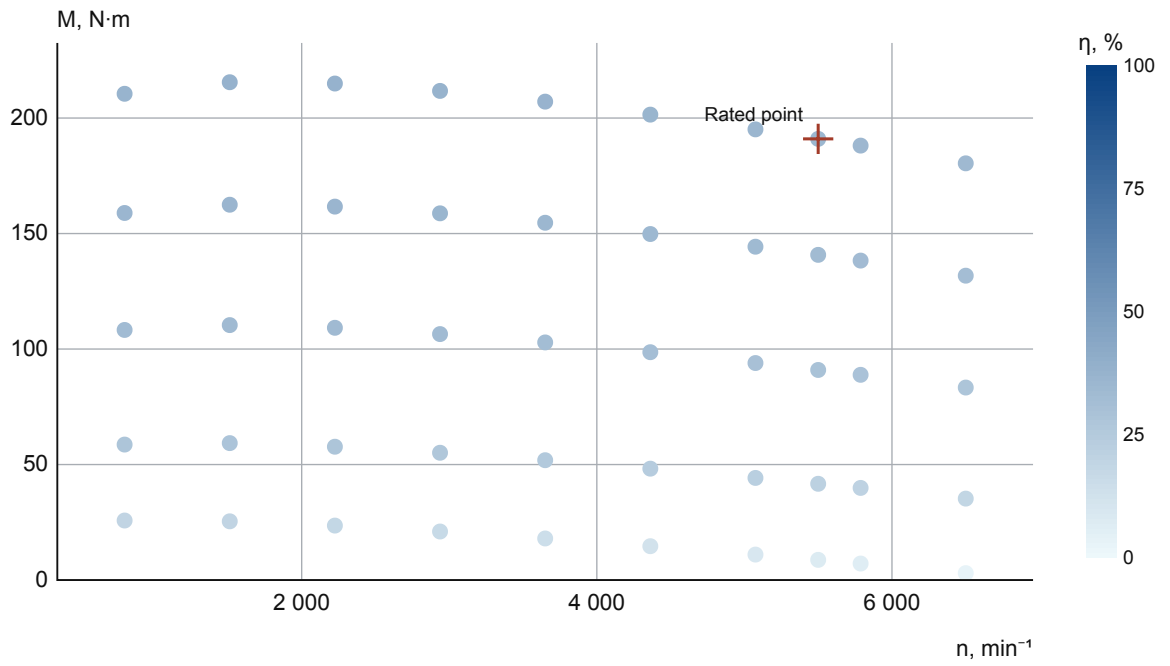


Figure 5.1. Full-load speed characteristic: power and torque.



Modelled points; the right scale shows efficiency η, %.

No interpolation across missing data. The cross marks the rated point.

Figure 5.2. Modelled engine brake-efficiency map versus speed and torque.

Revolutions per working cycle: $\mathrm{cycle}_{\mathrm{revolutions}}$ = 2 .

Crankshaft angular speed.

Crankshaft angular speed

where n_{req} — Speed at which power I-6 is required, s⁻¹

$$\omega_{\mathrm{e}} = 2 \cdot \pi \cdot n_{\mathrm{req}} \tag{5.1}$$

Substitute the current values in SI units:

$\omega_{\mathrm{e}} = 2 \cdot 3{,}142 \cdot 91{,}67 = 576 \ \mathrm{rad/s}$

The result passes to dependent calculations without being entered again.

Each cylinder completes a cycle in two revolutions for a four-stroke engine and one revolution for a two-stroke engine. The total cycle frequency converts J/cycle to W.

Working cycles of all cylinders per second

where z — Number of cylinders, 1; n_{req} — Speed at which power I-6 is required, s⁻¹; $\text{cycle}_{\text{revolutions}}$ — Revolutions per working cycle

$$\nu_{\text{c}} = \frac{z \cdot n_{\text{req}}}{\text{cycle}_{\text{revolutions}}} \tag{5.2}$$

Substitute the current values in SI units:

$$\nu_{\text{c}} = \frac{4 \cdot 91{,}67}{2} = 183{,}3 \ \text{s}^{-1}$$

The result passes to dependent calculations without being entered again.

Mean piston speed.

Mean piston speed

where S — Stroke adopted for subsequent calculations, m; n_{req} — Speed at which power I-6 is required, s⁻¹

$$c_{\text{m}} = 2 \cdot S \cdot n_{\text{req}} \tag{5.3}$$

Substitute the current values in SI units:

$$c_{\text{m}} = 2 \cdot 0{,}08727 \cdot 91{,}67 = 16 \ \text{m/s}$$

The result passes to dependent calculations without being entered again.

Indicated mean effective pressure.

Indicated mean effective pressure

where W_{i} — Indicated work per cylinder over 720°, J; V_{h} — Swept volume of one cylinder, m³

$$p_{\text{i}} = \frac{W_{\text{i}}}{V_{\text{h}}} \tag{5.4}$$

Substitute the current values in SI units:

$$p_{\text{i}} = \frac{694{,}9}{5{,}784 \times 10^{-4}} = 1201000 \ \text{Pa}$$

The result passes to dependent calculations without being entered again.

Engine indicated power.

Engine indicated power

where W_{i} — Indicated work per cylinder over 720°, J; ν_{c} — Working cycles of all cylinders per second, s⁻¹

$$P_{\mathrm{i}} = W_{\mathrm{i}} \cdot \nu_{\mathrm{c}} \tag{5.5}$$

Substitute the current values in SI units:

$$P_{\mathrm{i}} = 694{,}9 \cdot 183{,}3 = 127400 \ \mathrm{W}$$

The result passes to dependent calculations without being entered again.

Engine mechanical losses.

Engine mechanical losses

where p_{f} — Friction mean effective pressure, Pa; V_{h} — Swept volume of one cylinder, m³; ν_{c} — Working cycles of all cylinders per second, s⁻¹

$$P_{\mathrm{f}} = p_{\mathrm{f}} \cdot V_{\mathrm{h}} \cdot \nu_{\mathrm{c}} \tag{5.6}$$

Substitute the current values in SI units:

$$P_{\mathrm{f}} = 164100 \cdot 5{,}784 \times 10^{-4} \cdot 183{,}3 = 17400 \ \mathrm{W}$$

The result passes to dependent calculations without being entered again.

Modelled mechanical losses are deducted from indicated power. Only then is the result compared with the specified brake power.

Brake power after adopting the dimensions

where P_{i} — Engine indicated power, W; P_{f} — Engine mechanical losses, W

$$P_{\mathrm{ICE,calc}} = P_{\mathrm{i}} - P_{\mathrm{f}} \tag{5.7}$$

Substitute the current values in SI units:

$$P_{\mathrm{ICE,calc}} = 127400 - 17400 = 110000 \ \mathrm{W}$$

The result passes to dependent calculations without being entered again.

Brake mean effective pressure.

Brake mean effective pressure

where p_{i} — Indicated mean effective pressure, Pa; p_{f} — Friction mean effective pressure, Pa

$$p_{\mathrm{e}} = p_{\mathrm{i}} - p_{\mathrm{f}} \tag{5.8}$$

Substitute the current values in SI units:

$$p_{\mathrm{e}} = 1201000 - 164100 = 1037000 \ \mathrm{Pa}$$

The result passes to dependent calculations without being entered again.

Brake torque at the calculation point.

Brake torque at the calculation point

where $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W; ω_{e} — Crankshaft angular speed, rad/s

$$M_{\mathrm{e}} = \frac{P_{\mathrm{ICE,calc}}}{\omega_{\mathrm{e}}} \tag{5.9}$$

Substitute the current values in SI units:

$$M_{\mathrm{e}} = \frac{110000}{576} = 191 \ \mathrm{N{\cdot}m}$$

The result passes to dependent calculations without being entered again.

Fuel mass flow.

Fuel mass flow

where m_{fc} — Fuel mass per single-cylinder cycle, kg; ν_{c} — Working cycles of all cylinders per second, s⁻¹

$$m_{\mathrm{f}} = m_{\mathrm{fc}} \cdot \nu_{\mathrm{c}} \tag{5.10}$$

Substitute the current values in SI units:

$$m_{\mathrm{f}} = 3{,}847 \times 10^{-5} \cdot 183{,}3 = 0{,}007053 \ \mathrm{kg/s}$$

The result passes to dependent calculations without being entered again.

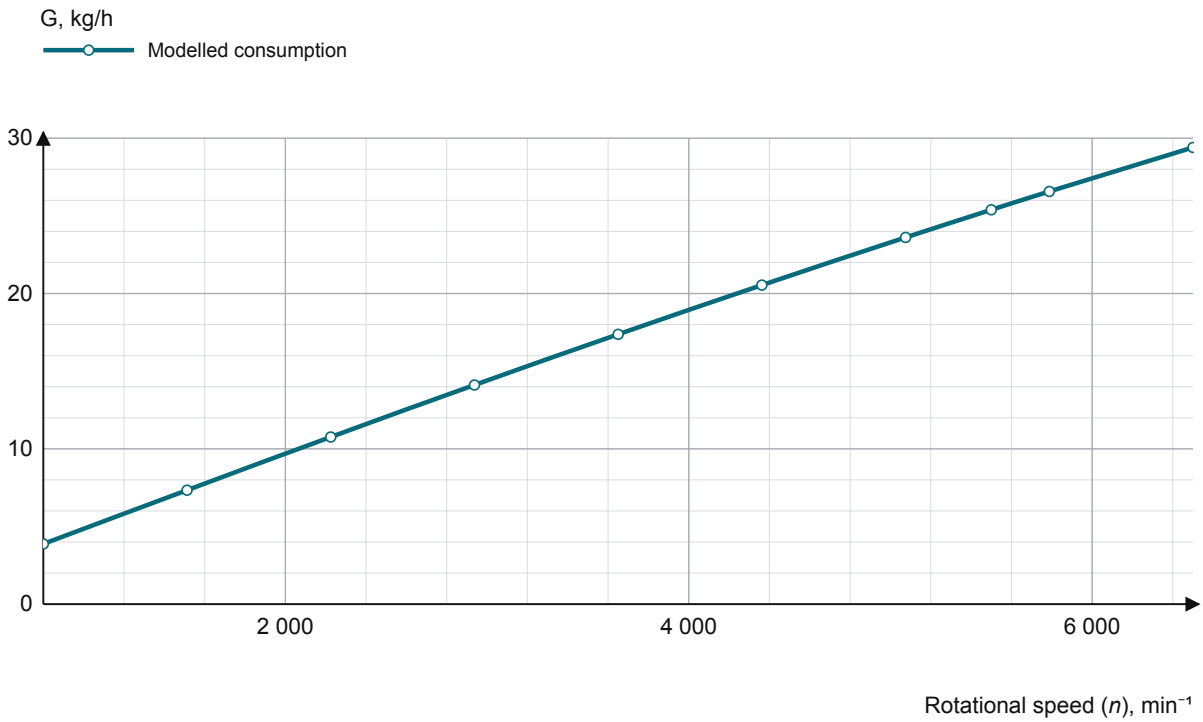


Figure 5.3. Hourly fuel consumption.

Fuel chemical-energy power.

Fuel chemical-energy power

where m_{f} — Fuel mass flow, kg/s; H_{u} — Lower heating value, J/kg

$$P_{\mathrm{H}} = m_{\mathrm{f}} \cdot H_{\mathrm{u}} \tag{5.11}$$

Substitute the current values in SI units:

$$P_{\mathrm{H}} = 0{,}007053 \cdot 4{,}4 \times 10^{7} = 310300 \ \mathrm{W}$$

The result passes to dependent calculations without being entered again.

Brake thermal efficiency — model estimate.

Brake thermal efficiency — model estimate

where $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W; P_{H} — Fuel chemical-energy power, W

$$\eta_{\mathrm{e}} = \frac{P_{\mathrm{ICE,calc}}}{P_{\mathrm{H}}} \tag{5.12}$$

Substitute the current values in SI units:

$$\eta_{\mathrm{e}} = \frac{110000}{310300} = 0{,}3545$$

The result passes to dependent calculations without being entered again.

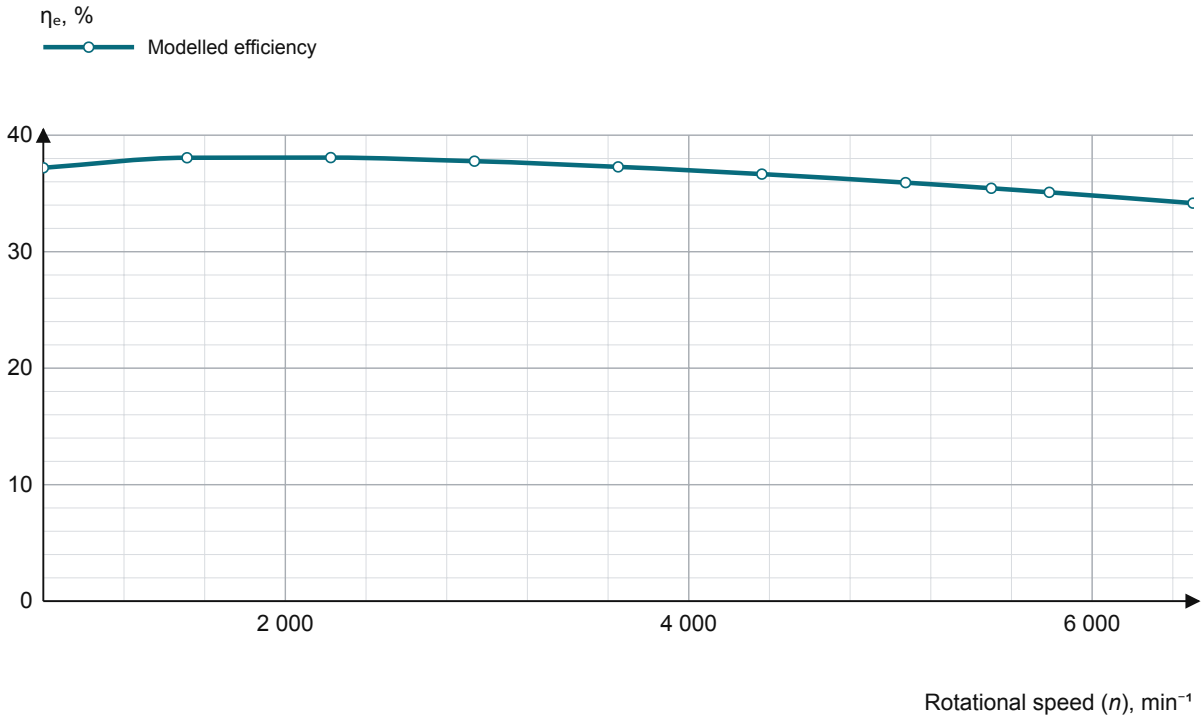


Figure 5.4. Engine brake thermal efficiency.

Brake specific fuel consumption in SI units.

Brake specific fuel consumption in SI units

where m_{f} — Fuel mass flow, kg/s; $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W

$$b_{\mathrm{e}} = \frac{m_{\mathrm{f}}}{P_{\mathrm{ICE,calc}}} \tag{5.13}$$

Substitute the current values in SI units:

$$b_{\mathrm{e}} = \frac{0{,}007053}{110000} = 6{,}412 \times 10^{-8}\ \mathrm{kg/J}$$

The result passes to dependent calculations without being entered again.

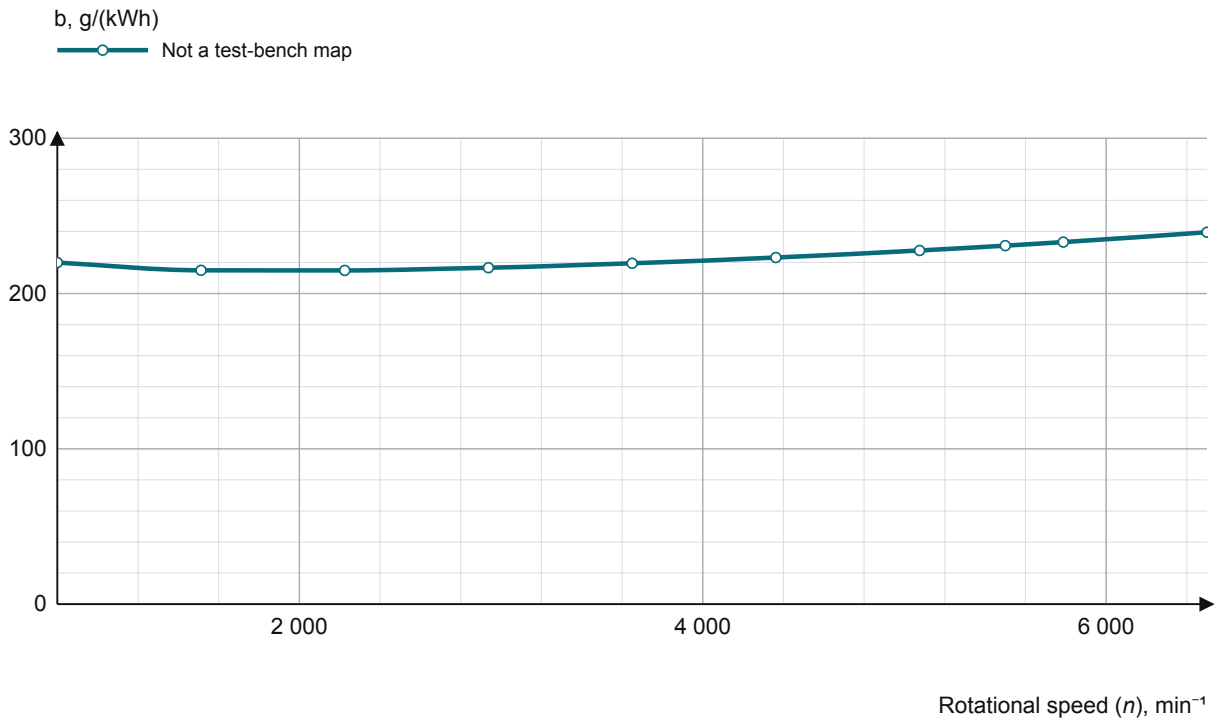


Figure 5.5. Modelled specific fuel consumption.

Volumetric fuel flow.

Volumetric fuel flow

where m_{f} — Fuel mass flow, kg/s; ρ_{f} — Fuel density, kg/m³

$$V_{\mathrm{f}} = \frac{m_{\mathrm{f}}}{\rho_{\mathrm{f}}} \tag{5.14}$$

Substitute the current values in SI units:

$$V_{\mathrm{f}} = \frac{0{,}007053}{745} = 9{,}467 \times 10^{-6}\ \mathrm{m^3/s}$$

Result in litres: V_{f} = 34,08 L/h.

The result passes to dependent calculations without being entered again.

The independently calculated power is compared with the requirement. The residual is not forced to zero by assigning the requirement to the result.

Relative engine requirement residual

where $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W; P_{req} — Required output-shaft power, W

$$\delta \mathrm{P}_{\mathrm{ICE}} = \frac{\left(P_{\mathrm{ICE,calc}} - P_{\mathrm{req}} \right)}{P_{\mathrm{req}}} \tag{5.15}$$

Substitute the current values in SI units:

$$\delta \mathrm{P}_{\mathrm{ICE}} = \frac{(110000 - 110000)}{110000} = 2{,}225 \times 10^{-6}$$

The result passes to dependent calculations without being entered again.

▼ Methodical mechanical-loss coefficients by chamber type

p_{f} = a_{f} + $c_{\mathrm{f}} \cdot c_{\mathrm{m}}$; pressure in MPa, speed in m/s [13]. For this linear correlation b_{f} = d_{f} = 0. Carburettor-engine coefficients are shown for comparison and are not automatically applied to PFI/GDI.

Table 5.2. Calculation results.

Design	a_{f}, MPa	c_{f}, MPa·s/m	Source equation
Carburettor, up to 6 cylinders, S/D > 1	0,049	0,0152	(1.50.1) [13]
Carburettor, 8 cylinders, S/D < 1	0,039	0,0132	(1.50.2) [13]
Carburettor, up to 6 cylinders, S/D < 1	0,034	0,0113	(1.50.3) [13]
Four-stroke diesel, undivided chamber	0,089	0,0118	(1.50.4) [13]
Pre-chamber diesel	0,103	0,0153	(1.50.5) [13]
Swirl-chamber diesel	0,089	0,0135	(1.50.6) [13]

Prescribed throttling p_{in}=p_{in0}(0.18+0.82L); fixed heat release.

Table 5.3. Calculation results.

n, min⁻¹	P_{e}, kW	M_{e}, N·m	b_{e}, g/(kWh)	p_{max}, MPa
800	17,64	210,5	219,9	7,04
1 513	34,13	215,5	214,9	7,105
2 225	50,08	214,9	214,9	7,116
2 938	65,14	211,8	216,6	7,094
3 650	79,16	207,1	219,5	7,054
4 363	92,04	201,5	223,2	7,002
5 075	103,7	195,1	227,7	6,942

n, min⁻¹	P_e, kW	M_e, N·m	b_e, g/(kWh)	p_{max}, MPa
5 500	110	191	230,8	6,902
5 788	114	188,1	233,1	6,874
6 500	122,8	180,4	239,5	6,796

6 Kinematics, dynamics, and auxiliary systems

Pressure and geometry from the preceding chapters determine gas, inertia, resultant, connecting-rod, and crank forces without manual copying. Total torque assumes evenly spaced firing events over one active cycle: 360°/z or 720°/z.

Energy fluctuation determines the required total inertia, not the final flywheel design. Heat flow gives a preliminary coolant mass flow, not pump head or a production pump. Strength, fatigue, spatial balancing, and bearings are not fully assessed here.

The additional 0.7 assessment uses current forces for preliminary axial/Euler/Goodman, piston-pin, and torsion checks, and heat for a series hydraulic circuit. A separate explicit specification is required; this is not a complete component or fatigue-life calculation. Unknown data are marked not assessed.

Table 6.1. Inputs.

Field	Adopted	Provenance
I-79 · Equivalent reciprocating mass per cylinder	0,59 kg	Given / adopted
I-80 · Allowable speed fluctuation	0,02	Given / adopted
I-81 · Coolant heat capacity	3 600 J/(kg·K)	Given / adopted
I-82 · Adopted coolant temperature rise	10 K	Given / adopted

Excess-energy fluctuation: ΔE = 380,8 J.

Maximum absolute connecting-rod force: S_{max} = 34,55 kN.

Maximum gas force: $F_{g,max}$ = 45,08 kN.

Maximum side force: N_{max} = 2,95 kN.

Maximum absolute inertia force: $F_{j,max}$ = 10,93 kN.

Pressure at the check point: p = 3,116 MPa.

Piston acceleration at the check point: a = 11 880 m/s².

Check-point crank angle: θ = 6,981 rad.

Connecting-rod angle at the check point: β = 0,181 rad.

The crank radius determines the tangential-force lever arm.

Crank radius

where S — Stroke adopted for subsequent calculations, m

$$R = \frac{S}{2} \tag{6.1}$$

Substitute the current values in SI units:

$$R = \frac{0{,}08727}{2} = 0{,}04364 \text{ m}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

Crankcase pressure is subtracted from cylinder pressure; absolute pressure alone does not give the gas force.

Gas force at the check point

where p — Pressure at the check point, Pa; p_{cr} — Absolute crankcase pressure, Pa; A_p — Piston area, m²

$$F_g = \left(p - p_{cr} \right) \cdot A_p \tag{6.2}$$

Substitute the current values in SI units:

$$F_g = \left(3116000 - 100000 \right) \cdot 0{,}006628 = 19990 \text{ N}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

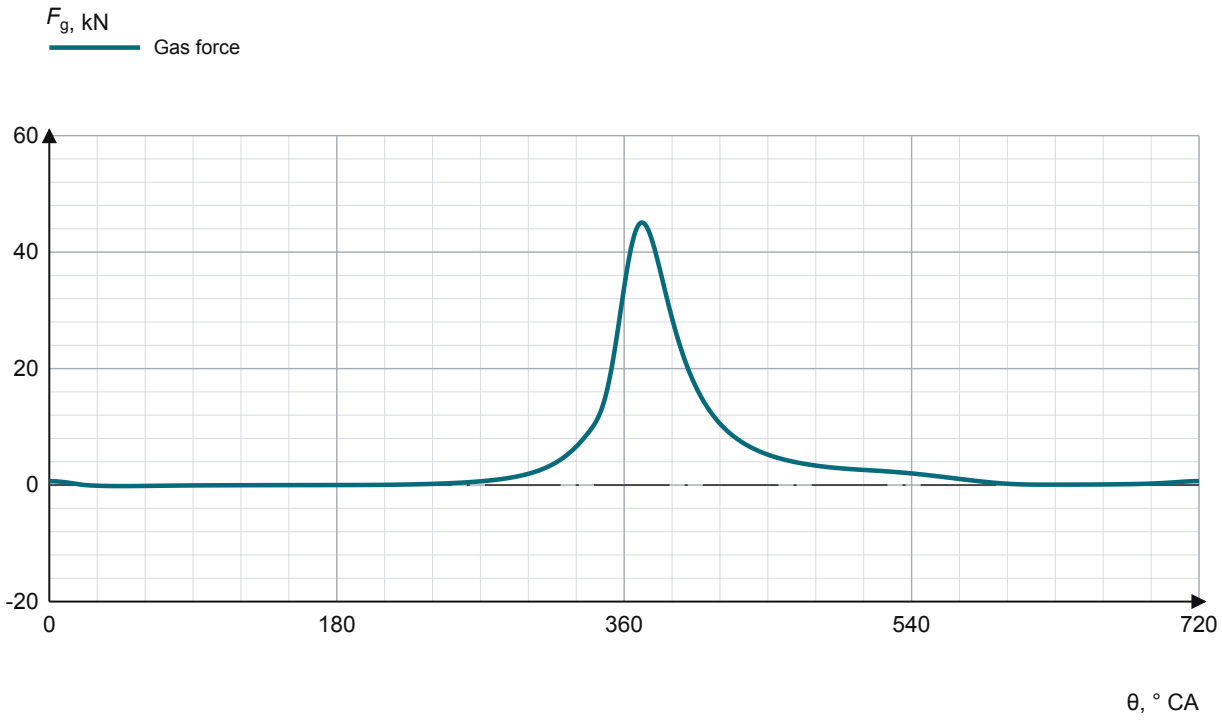


Figure 6.1. Gas force.

The inertia force acts opposite to piston acceleration.

Reciprocating-mass inertia force

where m_{rec} — Equivalent reciprocating mass per cylinder, kg; a — Piston acceleration at the check point, m/s²

$$F_{\text{j}} = -m_{\text{rec}} \cdot a \tag{6.3}$$

Substitute the current values in SI units:

$$F_{\text{j}} = -0{,}59 \cdot 11880 = \text{-}7012 \text{ N}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

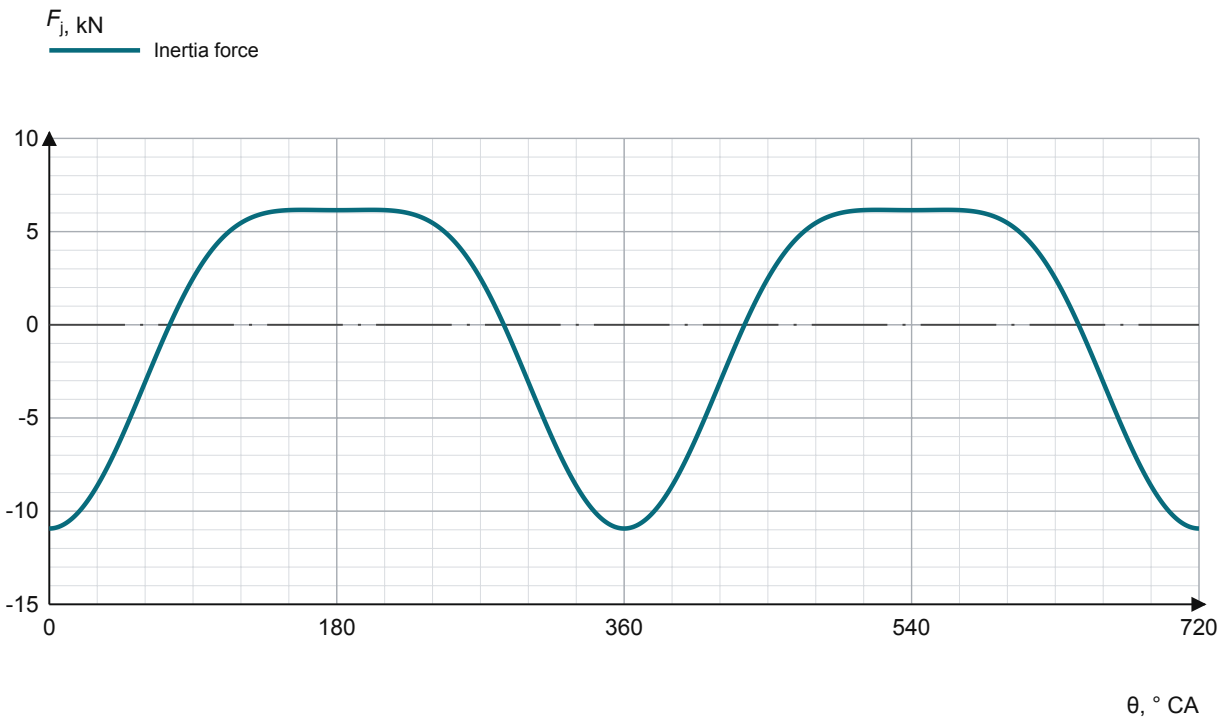


Figure 6.2. Inertia force.

The forces are summed algebraically using one sign convention, rather than adding their absolute magnitudes.

Resultant axial force

where F_{g} — Gas force at the check point, N; F_{j} — Reciprocating-mass inertia force, N

$$F_{\Sigma} = F_{\text{g}} + F_{\text{j}} \tag{6.4}$$

Substitute the current values in SI units:

$$F_{\Sigma} = 19990 + \text{-}7012 = 12980 \text{ N}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

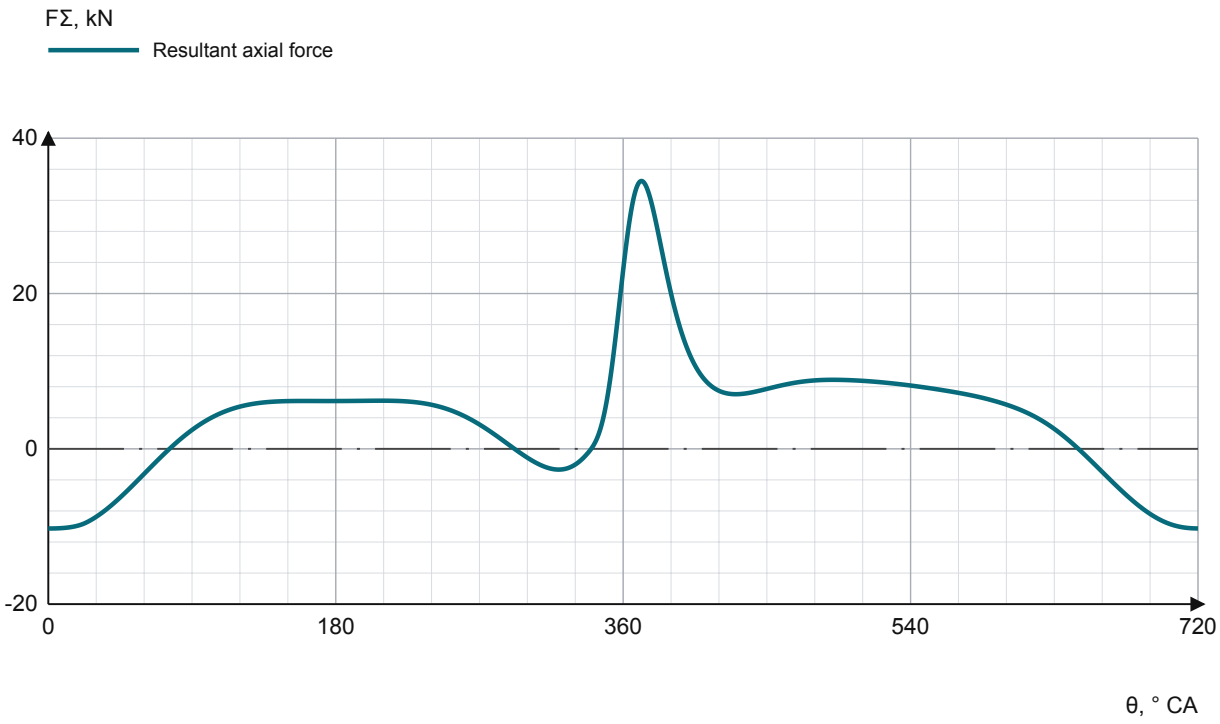


Figure 6.3. Resultant axial force.

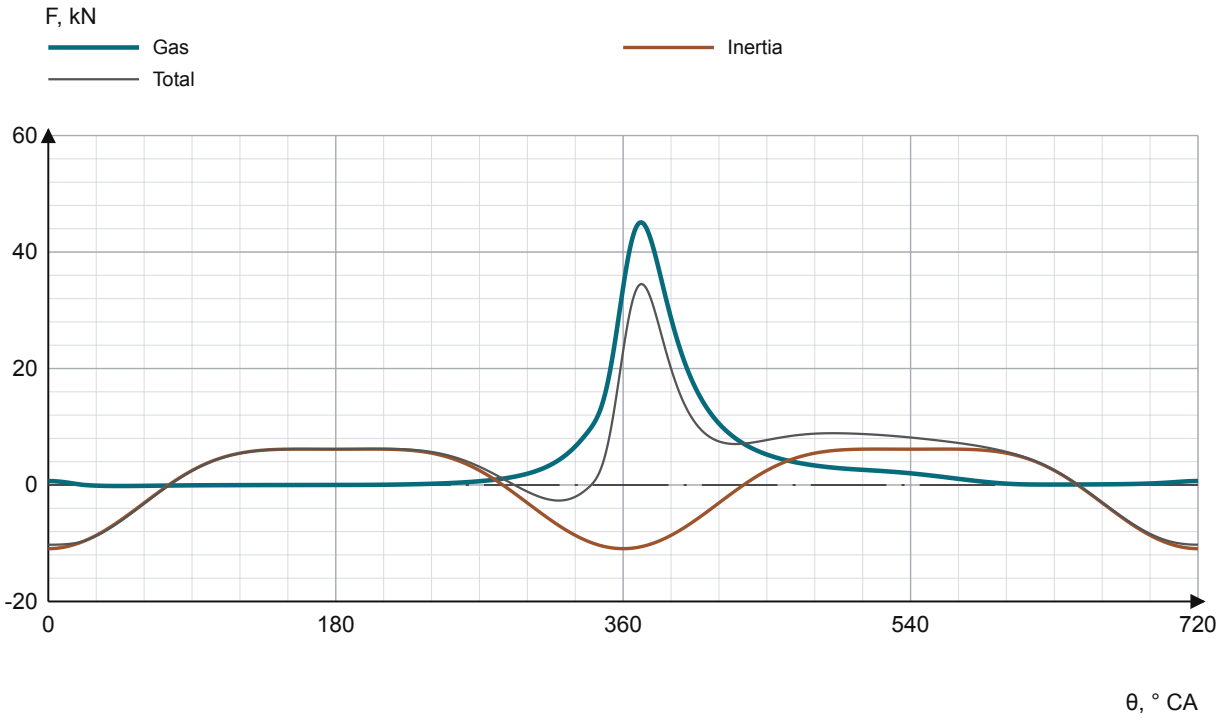


Figure 6.4. Axial forces of one cylinder.

The transverse component follows from the current connecting-rod angle.

Normal force on the cylinder wall

where F_{Σ} — Resultant axial force, N; β — Connecting-rod angle at the check point, rad

$$N_{\text{side}} = F_{\Sigma} \cdot \tan\left(\beta\right) \tag{6.5}$$

Substitute the current values in SI units:

$$N_{\text{side}} = 12980 \cdot \tan\left(0{,}181\right) = 2374 \text{ N}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

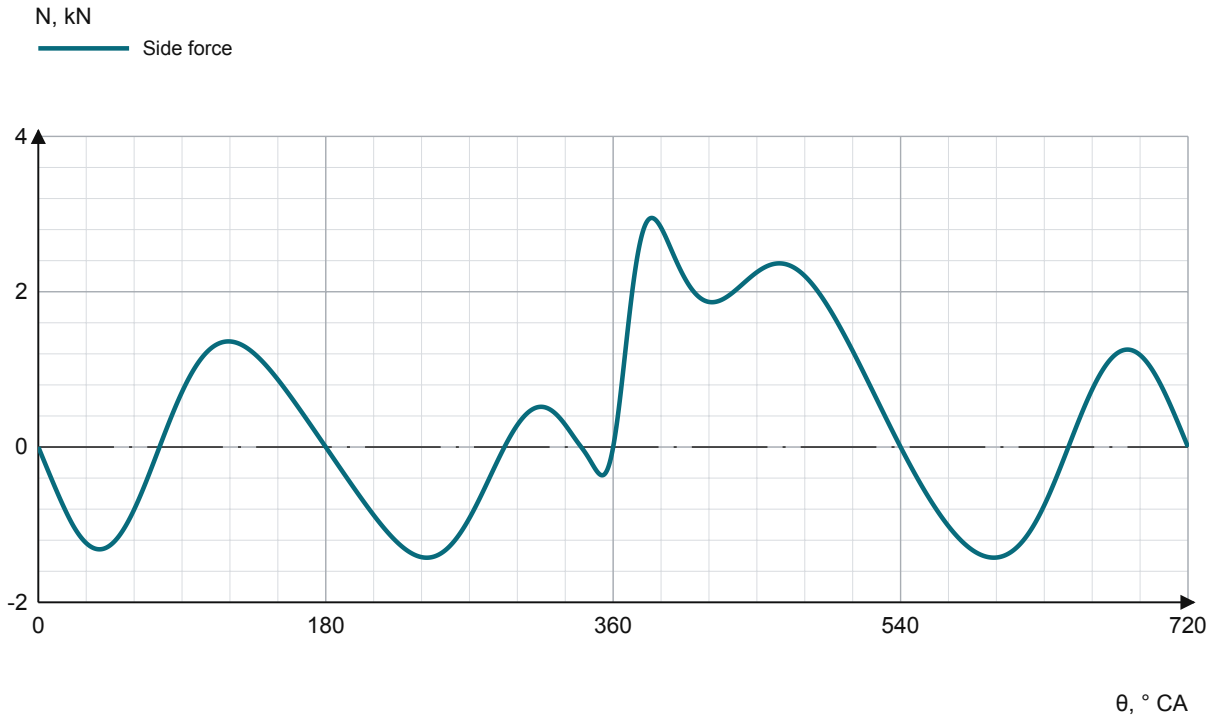


Figure 6.5. Side force.

Force along the connecting rod: FΣ/cos β. An inconsistent expression in the old RTF was not transferred.

Longitudinal force in the connecting rod

where F_{Σ} — Resultant axial force, N; β — Connecting-rod angle at the check point, rad

$$S_{\text{rod}} = \frac{F_{\Sigma}}{\cos\left(\beta\right)} \tag{6.6}$$

Substitute the current values in SI units:

$$S_{\text{rod}} = \frac{12980}{\cos\left(0{,}181\right)} = 13190 \text{ N}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

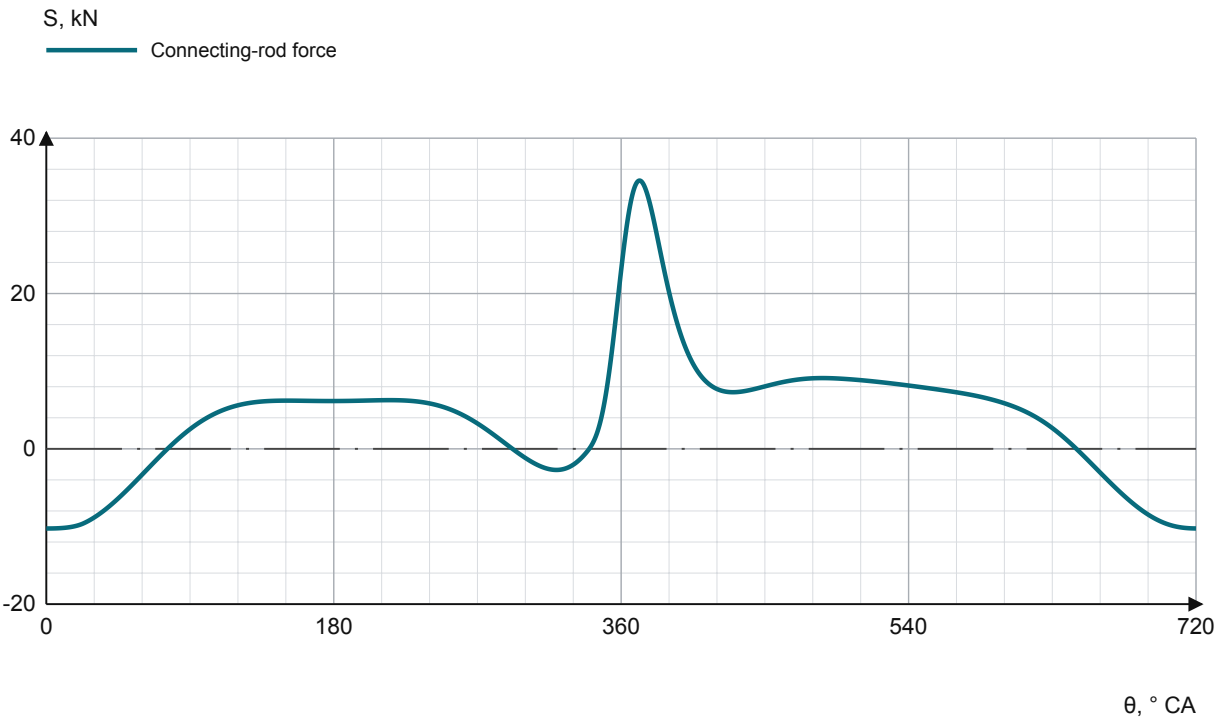


Figure 6.6. Connecting-rod force.

The tangential component produces torque; θ and β are passed as angles.

Crank tangential force

where F_Σ — Resultant axial force, N; θ — Check-point crank angle, rad; β — Connecting-rod angle at the check point, rad

$$T = \frac{F_\Sigma \cdot \sin\left(\theta + \beta\right)}{\cos\left(\beta\right)} \tag{6.7}$$

Substitute the current values in SI units:

$$T = \frac{12980 \cdot \sin\left(6{,}981 + 0{,}181\right)}{\cos\left(0{,}181\right)} = 10160 \text{ N}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

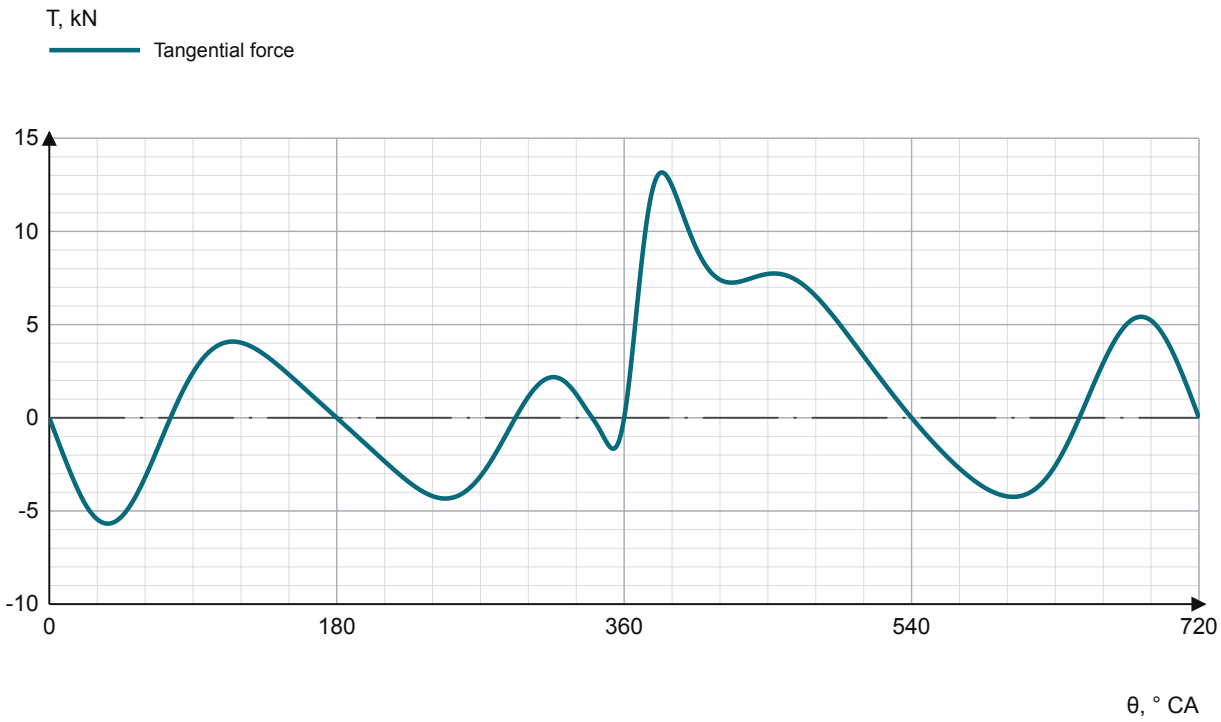


Figure 6.7. Tangential force.

The radial component loads the bearings; multiplying it by R does not give useful torque.

Crank radial force

where F_{Σ} — Resultant axial force, N; θ — Check-point crank angle, rad; β — Connecting-rod angle at the check point, rad

$$Z = \frac{F_{\Sigma} \cdot \cos\left(\theta + \beta\right)}{\cos\left(\beta\right)} \tag{6.8}$$

Substitute the current values in SI units:

$$Z = \frac{12980 \cdot \cos\left(6{,}981 + 0{,}181\right)}{\cos\left(0{,}181\right)} = 8414 \text{ N}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

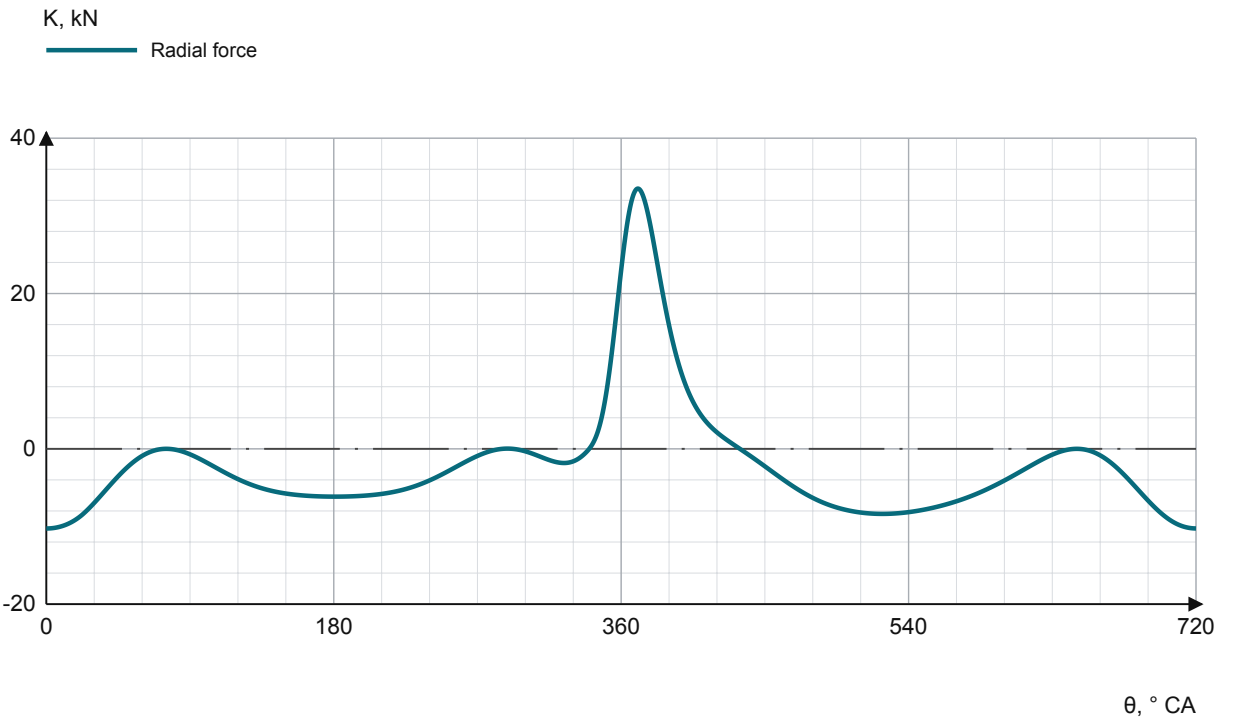


Figure 6.8. Radial force.

Torque is calculated from the tangential force and crank radius.

Single-cylinder torque at the check point

where T — Crank tangential force, N; R — Crank radius, m

$$M_{\mathrm{c}} = T \cdot R \tag{6.9}$$

Substitute the current values in SI units:

$$M_{\mathrm{c}} = 10160 \cdot 0{,}04364 = 443{,}3 \ \mathrm{N \cdot m}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

The integral of total torque over one active cycle equals the work of all cylinders. It is divided by 2πk rad, where k is the revolutions per cycle; mechanical losses have not yet been deducted.

Mean indicated torque of the complete engine

where W_{i} — Indicated work per cylinder over 720°, J; z — Number of cylinders, 1; $\mathrm{cycle}_{\mathrm{revolutions}}$ — Revolutions per working cycle

$$M_{\mathrm{i,avg}} = \frac{W_{\mathrm{i}} \cdot z}{\left(2 \cdot \pi \cdot \mathrm{cycle}_{\mathrm{revolutions}} \right)} \tag{6.10}$$

Substitute the current values in SI units:

$$M_{\mathrm{i,avg}} = \frac{694{,}9 \cdot 4}{\left(2 \cdot 3{,}142 \cdot 2 \right)} = 221{,}2 \ \mathrm{N \cdot m}$$

Do not confuse this with mean brake torque $M_e = P_e/\omega$, which already accounts for losses.

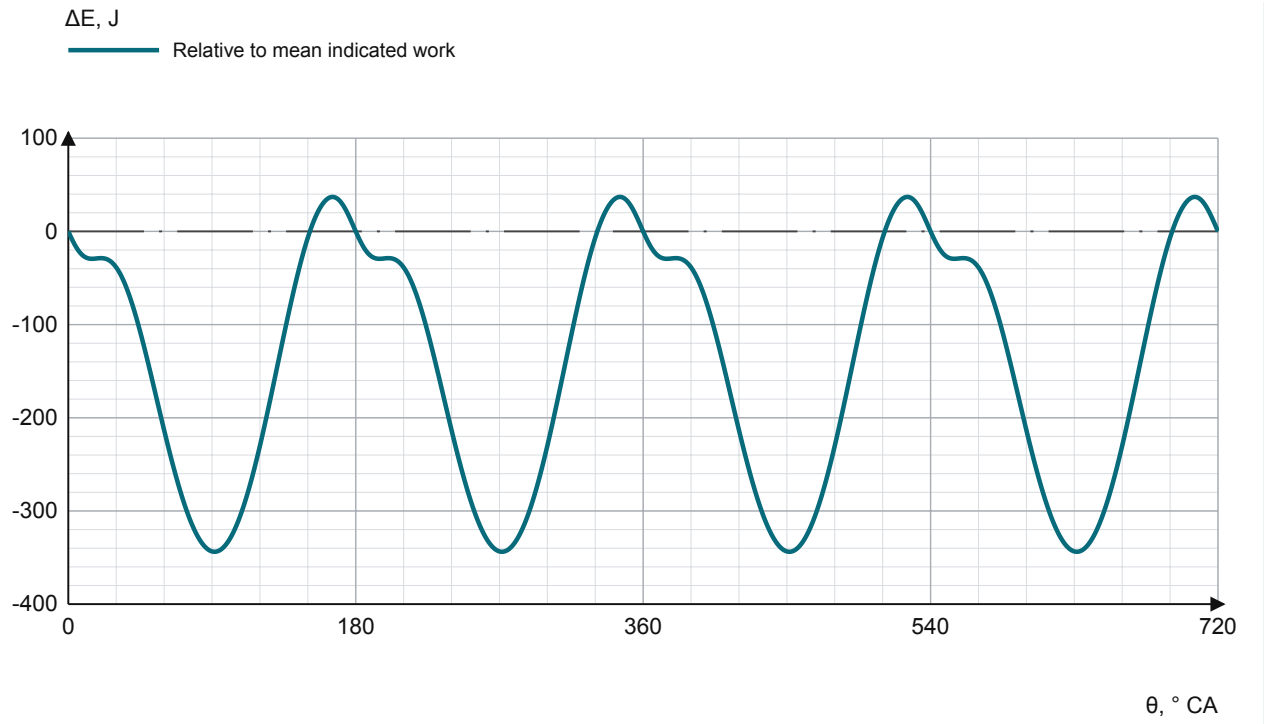


Figure 6.9. Excess-energy fluctuation.

The excess-energy fluctuation determines the required total shaft inertia. The inertia of other rotating components must be deducted separately before designing the flywheel.

Required total inertia referred to the shaft

where ΔE — Excess-energy fluctuation, J; δ_ω — Allowable speed fluctuation; ω_e — Crankshaft angular speed, rad/s

$$J_{req} = \frac{\Delta E}{\left(\delta_\omega \cdot {\omega_e}^2 \right)} \tag{6.11}$$

Substitute the current values in SI units:

$$J_{req} = \frac{380{,}8}{\left(0{,}02 \cdot 576^2 \right)} = 0{,}05739 \ \text{kg·m²}$$

The result passes to dependent calculations without being entered again.

Preliminary required coolant mass flow.

Preliminary required coolant mass flow

where Q_{w} — Heat flow to the walls, W; P_{f} — Engine mechanical losses, W; c_{c} — Coolant heat capacity, J/(kg·K); $\Delta \mathrm{T}_{\mathrm{c}}$ — Adopted coolant temperature rise, K

$$m_{\mathrm{c}} = \frac{\left(Q_{\mathrm{w}} + 0{,}7 \cdot P_{\mathrm{f}} \right)}{\left(c_{\mathrm{c}} \cdot \Delta \mathrm{T}_{\mathrm{c}} \right)} \tag{6.12}$$

Substitute the current values in SI units:

$$m_{\mathrm{c}} = \frac{\left(40870 + 0{,}7 \cdot 17400 \right)}{\left(3600 \cdot 10 \right)} = 1{,}474 \ \mathrm{kg/s}$$

The result passes to dependent calculations without being entered again.

Cylinder torque and power summation

For identical cylinders with evenly spaced firing: T = 720° for four-stroke or 360° for two-stroke; Δθ = T/i. Here T = 720°, i = 4, Δθ = 180°. Cyl 1…Cyl 4 identify phase positions, not a prescribed firing order of physical cylinders.

$M_j(\theta) = M_1((\theta - (j-1)\Delta\theta) \bmod T)$; $M\Sigma(\theta) = \Sigma_{j=1}^{i} M_j(\theta)$.

Phase shifts: 0°; 180°; 360°; 540°. Under these assumptions, the total curve repeats after Δθ and each cylinder curve after T. Uneven firing requires actual phase shifts.

$\omega = 2\pi n/60$; $P\Sigma(\theta) = \omega M\Sigma(\theta)$; $M_e = P_e/\omega$; $P_e = P_i - P_{мех.}$

At n = 5 500 min⁻¹: P_i = 127.4 kW; P_e = 110 kW; $M_{i,cep}$ = 221.2 N·m; M_e = 191 N·m. Instantaneous curves include gas forces and reciprocating-mass inertia before mechanical losses; the mean brake value is shown separately.

The full-load speed characteristic gives mean $P_e(n)$ and $M_e(n)$ of the complete engine at full load. This modelled characteristic uses fixed prescribed control laws; summation over crank angle θ shows pulsation at one speed n and does not replace the speed characteristic.

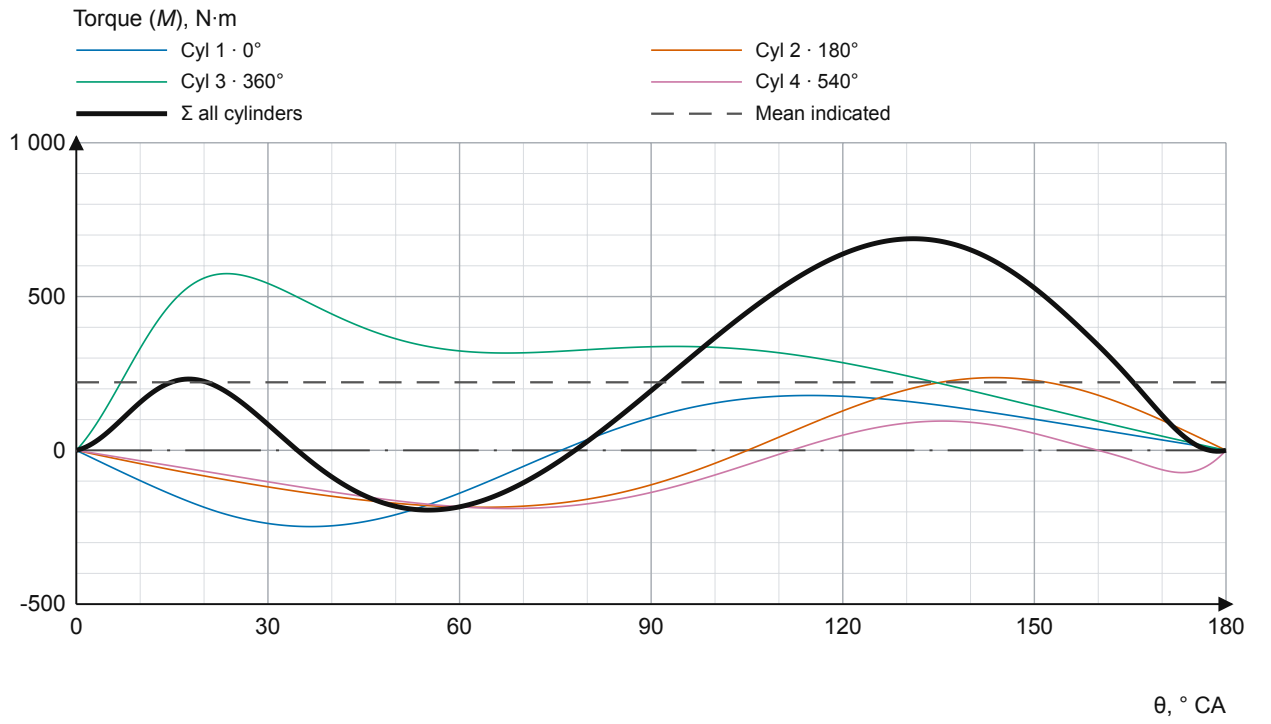


Figure 6.10. Cylinder contributions and total torque over one firing interval.

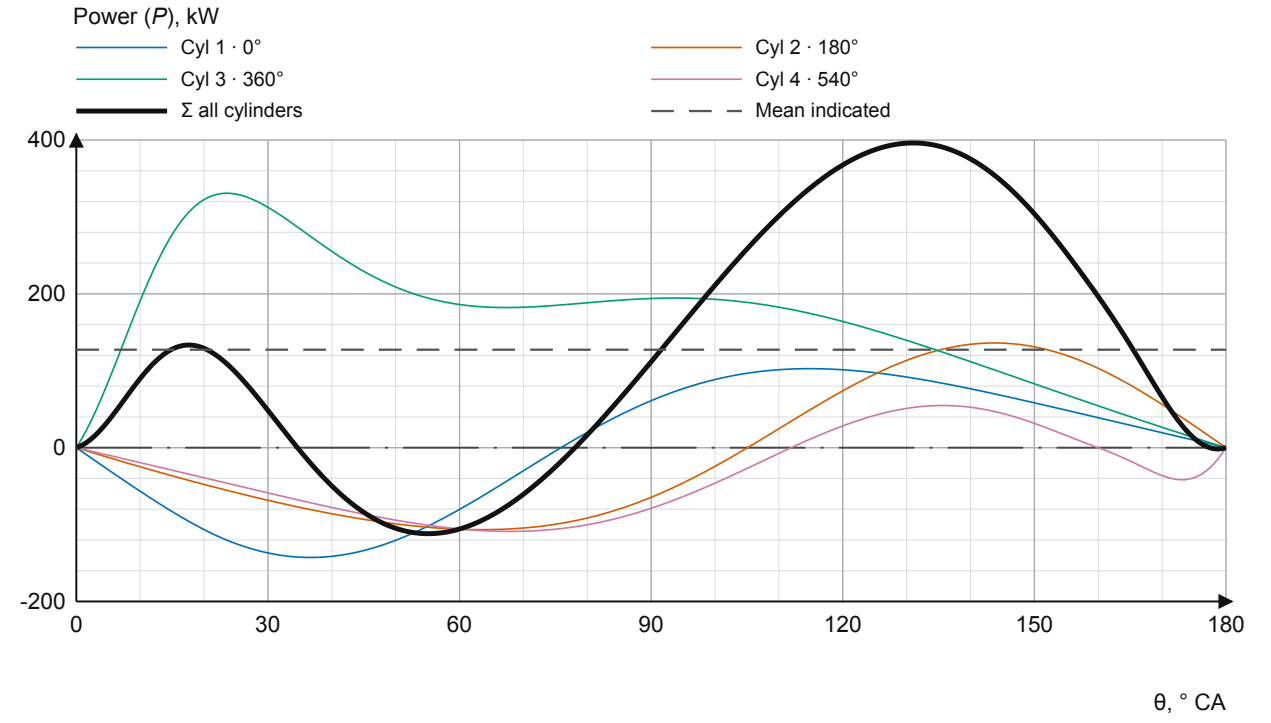


Figure 6.11. Cylinder contributions and total instantaneous power over one firing interval.

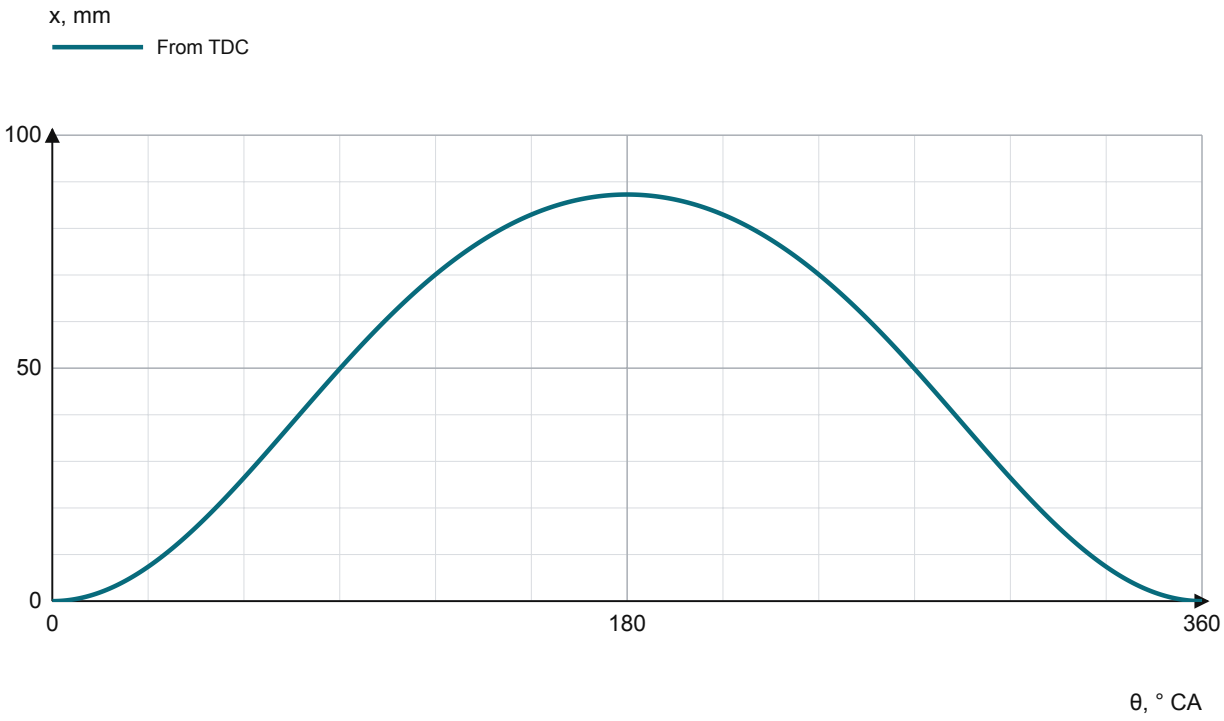


Figure 6.12. Piston displacement.

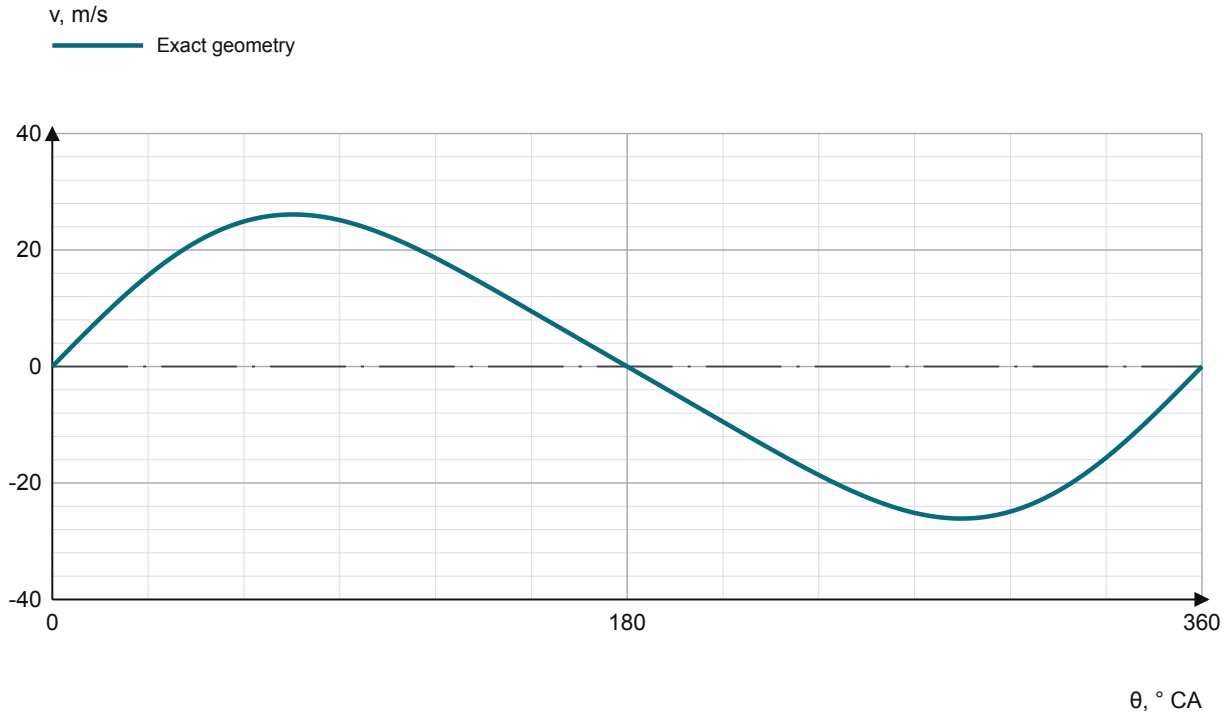


Figure 6.13. Piston speed.

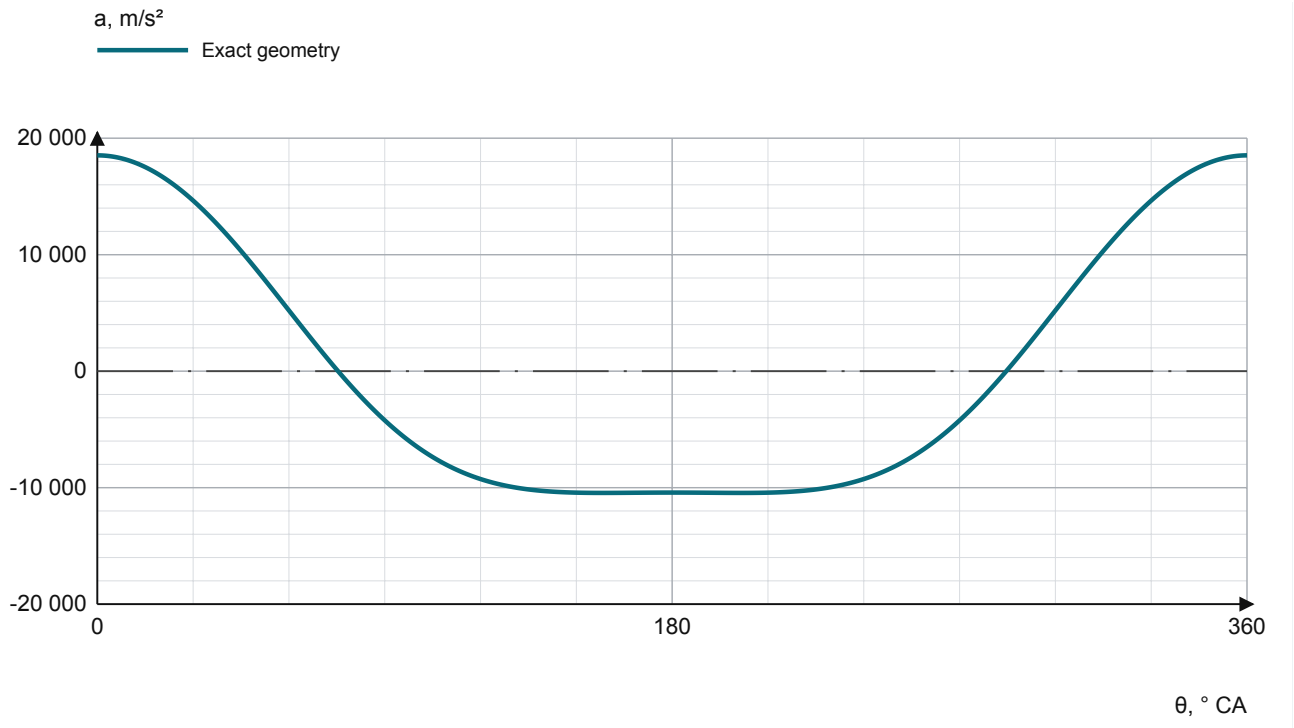


Figure 6.14. Piston acceleration.

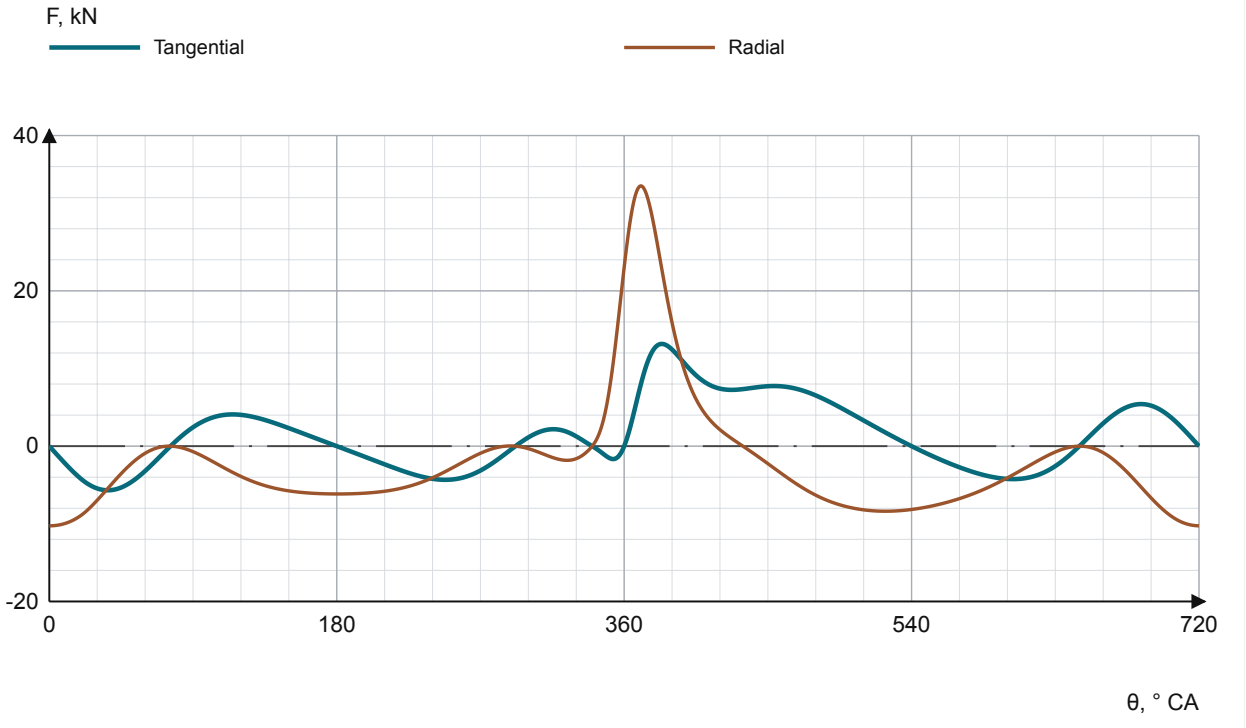


Figure 6.15. Crank loading.

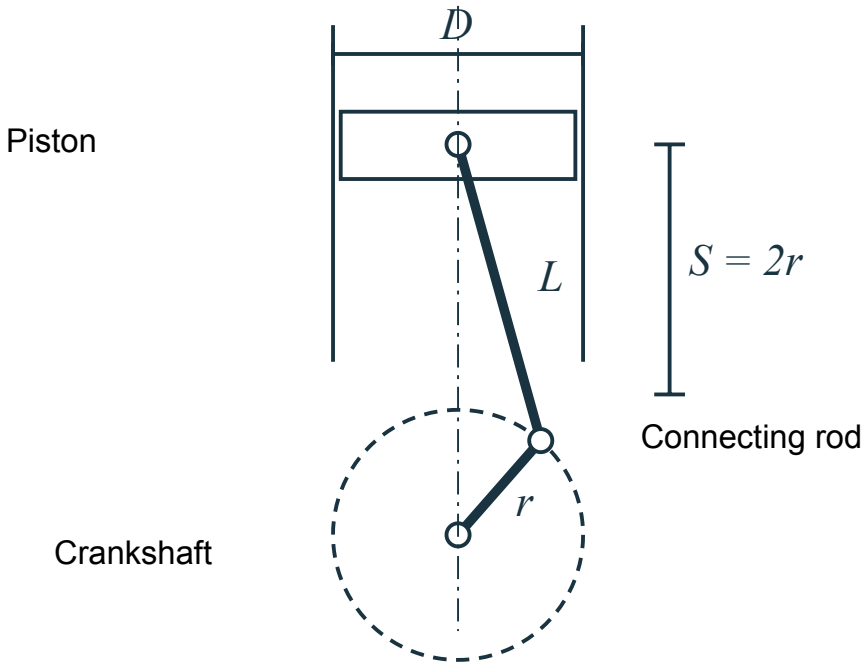


Figure 6.16. Single-cylinder kinematic diagram: D is bore; S is piston stroke; r is crank radius; L is rod length. Illustrative, not to scale.

11 EPA maps and driving scenarios

The US Environmental Protection Agency (EPA) publishes powertrain test-data packages and maps. In this chapter, EPA denotes the source of reference maps, not certification of this calculation.

A fuel-consumption map describes $\dot{m}f(n,M)$, not an indicator diagram. It does not replace the geometry, temperatures, or forces of the current engine cycle. Component identity, fuel, envelope, units, and missing values are retained [6].

Interpolation at the check point is restricted to the valid data domain. Empty and negative source cells do not become zero consumption. A map does not replace the design engine unless explicitly selected.

The current BEV vehicle and range are calculated in chapter 10. This chapter shows a separate external map for component comparison; it does not replace the calculated motor or complete vehicle.

The Bolt electric map represents motoring losses at 350 V. Inverter behaviour, a regeneration map, and verified continuous-current limits are not invented. It is used for the check point and plot; the general BEV continues to use the specified equivalent electric machine.

Table 11.1. Inputs.

Field	Adopted	Provenance
I-120 · Driving cycle: speed profile	WLTC Class 3b · 4 phases	Given / adopted [5]
I-121 · Engine-characteristic source for the driving cycle	Current design engine — no brand substitution	Given / adopted [6]
I-122 · Map check-point speed	2 000 min^{-1}	Given / adopted [6]
I-123 · Map check-point torque	100 N·m	Given / adopted [6]

Field	Adopted	Provenance
I-124 · Reference map for the point and plot	a25a-fks-tier3-202007	Given / adopted [6]

Mechanical power at the check point. [6]

Mechanical power at the check point

where n_{map} — Map check-point speed, s⁻¹; M_{map} — Map check-point torque, N·m

$$P_{\text{map}} = 2 \cdot \pi \cdot n_{\text{map}} \cdot M_{\text{map}} \tag{11.1}$$

Substitute the current values in SI units:

$$P_{\text{map}} = 2 \cdot 3{,}142 \cdot 33{,}33 \cdot 100 = 20940 \ \text{W}$$

A common check point is needed to convert the map consistently into efficiency and specific consumption.

Interpolated consumption of the specific component: mapFuel = 4,884 kg/h.

Lower heating value of the mapped fuel: mapLhv = 4,195 × 10⁷ J/kg.

Specific consumption from the EPA map. [6]

Specific consumption from the EPA map

where mapFuel — Interpolated consumption of the specific component, kg/s; P_{map} — Mechanical power at the check point, W

$$b_{\text{map}} = \frac{\text{mapFuel}}{P_{\text{map}}} \tag{11.2}$$

Substitute the current values in SI units:

$$b_{\text{map}} = \frac{0{,}001357}{20940} = 6{,}477 \times 10^{-8} \ \text{kg/J}$$

A missing map point has no numerical value and is not interpolated beyond the operating envelope.

Brake thermal efficiency at the EPA point. [6]

Brake thermal efficiency at the EPA point

where P_{map} — Mechanical power at the check point, W; mapFuel — Interpolated consumption of the specific component, kg/s; mapLhv — Lower heating value of the mapped fuel, J/kg

$$\eta_{\text{map}} = \frac{P_{\text{map}}}{(\ \text{mapFuel} \cdot \text{mapLhv}\ )} \tag{11.3}$$

Substitute the current values in SI units:

$$\eta_{\text{map}} = \frac{20940}{\left(\ 0{,}001357 \cdot 4{,}195 \times 10^{7}\ \right)} = 0{,}368$$

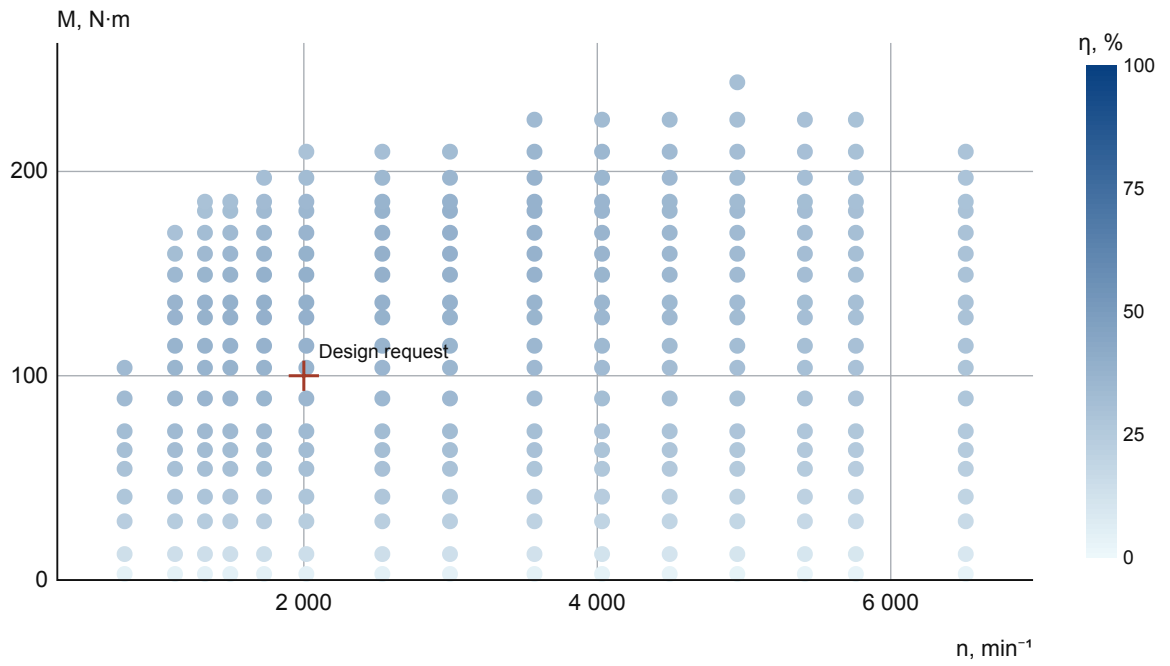
The fuel heating value comes from the specific map package, not from another engine's fuel.

Reference map: epa-a25a-fks-tier3-202007. Reference component only, does not replace the current engine cycle. 273 valid nodes; values are not extrapolated outside the data domain.

Request: n=2 000 min⁻¹; M=100 N·m. The point lies within the valid map domain.

The driving example is disabled. Select Target vehicle: yes in chapter 10. The reference map above remains available independently.

UDDS/HWFET/US06 here are supplied library profile copies. Checksums verify local data integrity, not a byte-for-byte comparison with current EPA originals. Tables and masks are retained in Data/EPA.



Known external-component points; the right-hand scale shows efficiency η, %.

No interpolation across missing data. The cross marks the design request.

Figure 11.1. External-component efficiency map: known points and design request.

12 Strength and fatigue life: first approximation

Loads come from the current cycle. The connecting rod is a bar; the pin is a simply supported beam; the shaft is a circular section in torsion. Default materials, concentration factors, and geometry are illustrative and editable.

Goodman uses the complete block's stress amplitude and mean; no benefit from compressive mean stress is credited. Euler applies only when critical stress is below the specified elastic limit. These are preliminary checks, not a spatial crankshaft analysis [12].

S–N: logarithmic interpolation between two entered reference points. One block repetition is 360° or 720° according to the selected cycle. Life is not extrapolated beyond the S–N interval; only a bound or insufficient data is stated. There is no automatically verified material fatigue curve, rainflow counting, or independent fatigue-life validation.

Table 12.1. Inputs.

Field	Adopted	Provenance
I-125 · Connecting-rod section area	225 mm²	Given / adopted
I-126 · Weak-section second moment of area	750 mm⁴	Given / adopted
I-127 · Young's modulus	210 000 MPa	Given / adopted
I-128 · Component yield strength	600 MPa	Given / adopted
I-129 · Ultimate tensile strength	800 MPa	Given / adopted
I-130 · Adopted component endurance limit	250 MPa	Given / adopted

Field	Adopted	Provenance
I-131 · Piston-pin outer diameter	28 mm	Given / adopted
I-132 · Piston-pin inner diameter	14 mm	Given / adopted
I-133 · Calculated piston-pin span	40 mm	Given / adopted
I-134 · Projected contact length	25 mm	Given / adopted
I-135 · Equivalent shaft-section diameter	45 mm	Given / adopted
I-136 · Allowable shear stress	150 MPa	Given / adopted
I-137 · Allowable mean piston-pin pressure	100 MPa	Given / adopted
I-138 · S–N amplitude at N_1	500 MPa	Given / adopted
I-139 · S–N amplitude at N_2	200 MPa	Given / adopted
I-140 · Concentration / fatigue factor	1,4	Given / adopted
I-141 · Effective-length factor	1	Given / adopted
I-142 · Reference cycle count N_1	100 000	Given / adopted
I-143 · Reference cycle count N_2	1×10^7	Given / adopted

Maximum connecting-rod compression force over 720°: rodMax = 34,55 kN.

Minimum connecting-rod compression force over 720°: rodMin = -10,26 kN.

Nominal connecting-rod stress amplitude. [12]

Nominal connecting-rod stress amplitude

where rodMax — Maximum connecting-rod compression force over 720°, N; rodMin — Minimum connecting-rod compression force over 720°, N; A_{rod} — Connecting-rod section area, m²

$$\sigma_{\mathrm{a}} = \frac{(\,\mathrm{rodMax} - \mathrm{rodMin}\,)}{(\,2 \cdot A_{\mathrm{rod}}\,)} \tag{12.1}$$

Substitute the current values in SI units:

$$\sigma_{\mathrm{a}} = \frac{(\,34550 - \text{-}10260\,)}{\left(\,2 \cdot 2{,}25 \times 10^{-4}\,\right)} = 9{,}957 \times 10^{7}\ \mathrm{Pa}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Mean axial stress, positive in tension. [12]

Mean axial stress, positive in tension

where rodMax — Maximum connecting-rod compression force over 720°, N; rodMin — Minimum connecting-rod compression force over 720°, N; A_{rod} — Connecting-rod section area, m²

$$\sigma_{\mathrm{m}} = \frac{-\left(\mathrm{rodMax} + \mathrm{rodMin}\right)}{\left(2 \cdot A_{\mathrm{rod}}\right)} \tag{12.2}$$

Substitute the current values in SI units:

$$\sigma_{\mathrm{m}} = \frac{-\left(34550 + \text{-}10260\right)}{\left(2 \cdot 2{,}25 \times 10^{-4}\right)} = \text{-}5{,}398 \times 10^{7}\ \mathrm{Pa}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Tensile part of mean stress: $\mathrm{tensileMean}$ = 0 MPa.

Goodman endurance utilisation. [12]

Goodman endurance utilisation

where K_{f} — Concentration / fatigue factor; σ_{a} — Nominal connecting-rod stress amplitude, Pa; S_{e} — Adopted component endurance limit, Pa; $\mathrm{tensileMean}$ — Tensile part of mean stress, Pa; S_{u} — Ultimate tensile strength, Pa

$$U_{\mathrm{G}} = \frac{K_{\mathrm{f}} \cdot \sigma_{\mathrm{a}}}{S_{\mathrm{e}}} + \frac{K_{\mathrm{f}} \cdot \mathrm{tensileMean}}{S_{\mathrm{u}}} \tag{12.3}$$

Substitute the current values in SI units:

$$U_{\mathrm{G}} = \frac{1{,}4 \cdot 9{,}957 \times 10^{7}}{2{,}5 \times 10^{8}} + \frac{1{,}4 \cdot 0}{8 \times 10^{8}} = 0{,}5576$$

$U_{\mathrm{G}} \le 1$ passes only this simplified criterion. Mean compressive stress is taken as zero for this criterion.

Euler critical load of the weak section. [12]

Euler critical load of the weak section

where E — Young's modulus, Pa; I_{rod} — Weak-section second moment of area, m⁴; K_{L} — Effective-length factor; l — Connecting rod centre distance, m

$$F_{\mathrm{E}} = \frac{\pi^{2} \cdot E \cdot I_{\mathrm{rod}}}{\left(K_{\mathrm{L}} \cdot l\right)^{2}} \tag{12.4}$$

Substitute the current values in SI units:

$$F_{\mathrm{E}} = \frac{3{,}142^{2} \cdot 2{,}1 \times 10^{11} \cdot 7{,}5 \times 10^{-10}}{\left(1 \cdot 0{,}1558\right)^{2}} = 64010\ \mathrm{N}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Second moment of area of the piston-pin section. [12]

Second moment of area of the piston-pin section

where d_{p} — Piston-pin outer diameter, m; d_{pi} — Piston-pin inner diameter, m

$$I_{\mathrm{p}} = \frac{\pi \cdot \left({d_{\mathrm{p}}}^{4} - {d_{\mathrm{pi}}}^{4} \right)}{64} \tag{12.5}$$

Substitute the current values in SI units:

$$I_{\mathrm{p}} = \frac{3{,}142 \cdot \left(0{,}028^{4} - 0{,}014^{4} \right)}{64} = 2{,}829 \times 10^{-8}\ \mathrm{m^4}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Piston-pin bending stress: central load. [12]

Piston-pin bending stress: central load

where K_{f} — Concentration / fatigue factor; $S_{\max}$ — Maximum absolute connecting-rod force, N; l_{p} — Calculated piston-pin span, m; d_{p} — Piston-pin outer diameter, m; I_{p} — Second moment of area of the piston-pin section, m⁴

$$\sigma_{\mathrm{p}} = \frac{K_{\mathrm{f}} \cdot S_{\max} \cdot l_{\mathrm{p}} \cdot d_{\mathrm{p}}}{\left(8 \cdot I_{\mathrm{p}} \right)} \tag{12.6}$$

Substitute the current values in SI units:

$$\sigma_{\mathrm{p}} = \frac{1{,}4 \cdot 34550 \cdot 0{,}04 \cdot 0{,}028}{\left(8 \cdot 2{,}829 \times 10^{-8} \right)} = 2{,}394 \times 10^{8}\ \mathrm{Pa}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Mean projected piston-pin pressure. [12]

Mean projected piston-pin pressure

where $S_{\max}$ — Maximum absolute connecting-rod force, N; d_{p} — Piston-pin outer diameter, m; b_{p} — Projected contact length, m

$$p_{\mathrm{p}} = \frac{S_{\max}}{\left(d_{\mathrm{p}} \cdot b_{\mathrm{p}} \right)} \tag{12.7}$$

Substitute the current values in SI units:

$$p_{\mathrm{p}} = \frac{34550}{\left(0{,}028 \cdot 0{,}025 \right)} = 4{,}935 \times 10^{7}\ \mathrm{Pa}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Maximum absolute total indicated torque: $M_{\mathrm{i\Sigma,max}}$ = 688 N·m.

Torsion of an equivalent solid section. [12]

Torsion of an equivalent solid section

where K_f — Concentration / fatigue factor; $M_{i\Sigma,max}$ — Maximum absolute total indicated torque, N·m; d_s — Equivalent shaft-section diameter, m

$$\tau_s = \frac{16 \cdot K_f \cdot M_{i\Sigma,max}}{\left(\pi \cdot d_s^{\,3}\right)} \tag{12.8}$$

Substitute the current values in SI units:

$$\tau_s = \frac{16 \cdot 1{,}4 \cdot 688}{\left(3{,}142 \cdot 0{,}045^3\right)} = 5{,}383 \times 10^7 \text{ Pa}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Table 12.2. Requirement checks.

Preliminary check	State	Value / limit
Connecting rod: illustrative Goodman criterion	Passed under the assumptions	0,5576 / 1
Connecting rod: Euler only within the elastic domain	Passed under the assumptions	34,55 kN / 64,01 kN
Piston pin: illustrative bending	Passed under the assumptions	239,4 MPa / 600 MPa
Piston pin: mean pressure	Passed under the assumptions	49,35 MPa / 100 MPa
Equivalent shaft: torsion only	Passed under the assumptions	53,83 MPa / 150 MPa

Euler applies only when σcr = Fcr/A ≤ 0.6·σy. The factor 0.6 is a conservative assumption of this educational approximation, not a manufacturer's proportional limit. Beyond it, the result is not accepted as a buckling check. The shaft here is straight and circular, without crank webs, fillets, or bending.

$$\sigma_{eq} = K_f \cdot \sigma_a \,/\, \left(1 - K_f \cdot \max\left(\sigma_m, 0\right) \,/\, \sigma_u\right) \tag{12.9}$$

$\sigma_{eq} = 1{,}4 \cdot 99{,}57 \,/\, \left(1 - 1{,}4 \cdot 0 \,/\, 800\right) = 139{,}4\text{МПа}.$

$$b = \ln\left(\sigma_2 / \sigma_1\right) / \ln\left(N_2 / N_1\right)$$
$$N = N_1 \cdot \left(\sigma_{eq} / \sigma_1\right)^{1/b} \tag{12.10}$$

$b = \ln\left(0{,}4\right) / \ln\left(100\right) = \text{-}0{,}199;$ інтерполяція N за цим опорами не застосовна.

Illustrative fatigue life: stress is below the lower S–N reference point; N > 1 × 10^7 only under a monotonicity assumption. This is not infinite life. This represents one equivalent repeated block, not a rainflow spectrum. The S–N values are illustrative; actual life is not validated without material data and measurements.

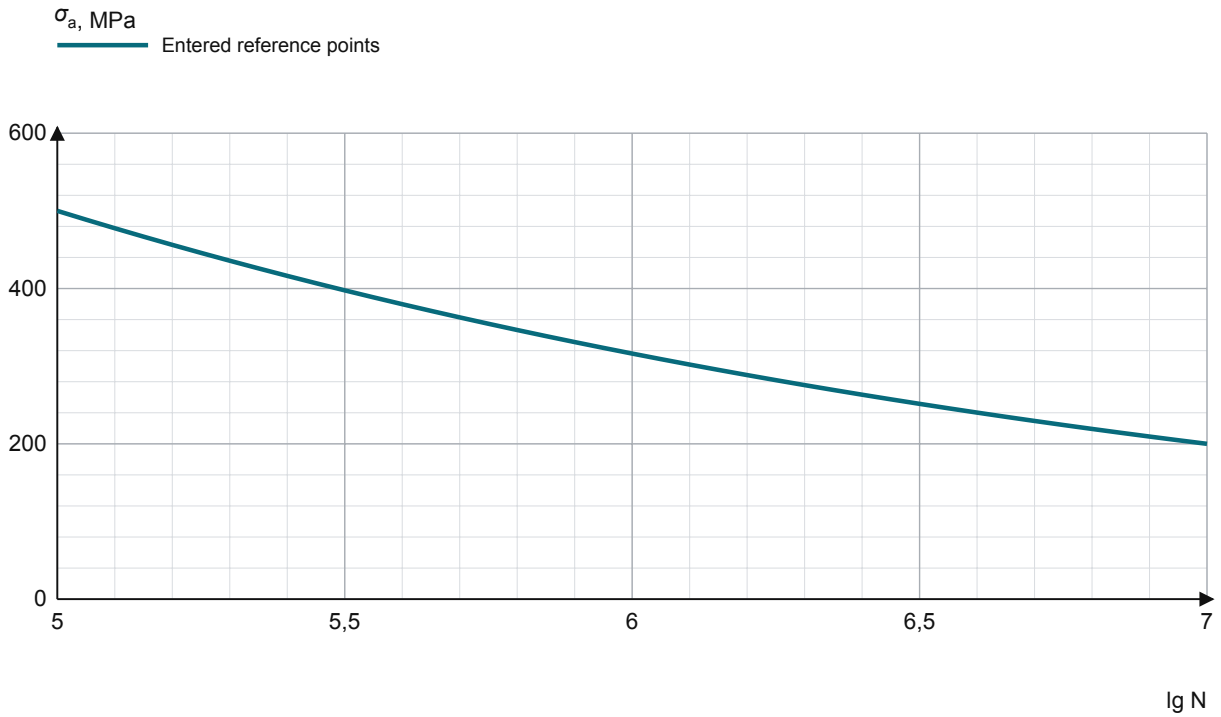


Figure 12.1. Illustrative S–N curve: only between entered reference points.

13 Simplified control strategy

A first-order actuator is added to the traction source: τ dP/dt + P = P_{cmd}. An exact discrete solution for a constant command is used; τ and load fraction are editable educational assumptions. The transient plot is a separate check and does not modify current engine-cycle pressure.

The driving cycle uses the Powertrain limiting strategy: traction, battery charging/discharging, regeneration, P2, or series generation. Missing map points prevent reliable fuel-consumption calculation rather than disappearing from the balance. This algorithm is not ECU/BMS firmware.

Table 13.1. Inputs.

Field	Adopted	Provenance
I-144 · τ_{ctl} — Actuator-response time constant	0,35 s	Given / adopted
I-145 · Δt_{ctl} — Check-transient time step	0,02 s	Given / adopted
I-146 · t_{ctl} — Response-check duration	3 s	Given / adopted
I-147 · Requested load fraction	0,8	Given / adopted

Required power command. [12]

> **Required power command**
>
> where u — Requested load fraction; P_{calc} — Available output-shaft power, W
>
> $$P_{\text{cmd}} = u \cdot P_{\text{calc}} \tag{13.1}$$
>
> Substitute the current values in SI units:
>
> $P_{\text{cmd}} = 0{,}8 \cdot 110000 = 88000\ \text{W}$
>
> This is an initial estimate under explicit assumptions, not manufacturing approval.

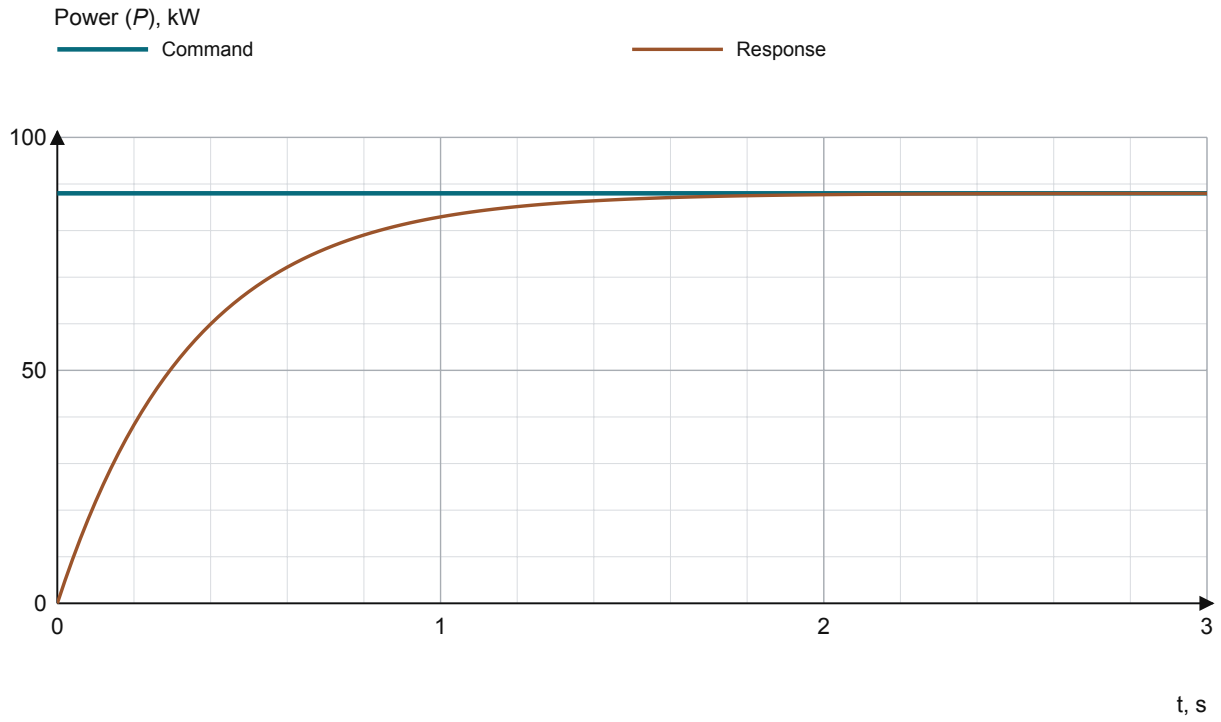


Figure 13.1. First-order power response.

$$\begin{gathered} \tau \cdot \mathrm{dP} \,/\, \mathrm{dt} + P = P_{\text{cmd}} \\ P(\,t\,) = P_{\text{cmd}} \cdot (\,1 - \exp(\,-t \,/\, \tau\,)\,) \end{gathered} \tag{13.2}$$

$P(\,3\text{s}\,) = 88 \cdot (\,1 - \exp(\,-3 \,/\, 0{,}35\,)\,) = 87{,}98\text{кВт.Початково}P(\,0\,) = 0; \tau = 0{,}35\text{s}.$

Numerical step Δt=0.02 s; exact transition for a constant command. Residual from the analytical solution is 0 W. This test demonstrates actuator lag; it does not change quasi-static EPA maps and is not presented as an ECU.

The hybrid driving example separately uses a simplified power split based on demand and SOC, including current and regeneration limits. Cold starts, catalysts, knock, and manufacturer control maps are not included.

15 Comparison with measurements

Paired CSV import accepts observed and predicted values with time, trip, and units. RMSE, bias, and maximum absolute error are calculated only from supplied pairs. Empty cells are excluded and counted separately.

Training and validation trips do not overlap. Until the user's measurements are supplied, the status is Not experimentally validated. The demonstration CSV checks arithmetic, not engine accuracy. The user's OBD2 data are not mixed with independent torque measurements.

> No validation CSV was supplied for this calculation. Independent experimental verification has not been performed. A software test using synthetic pairs checks only metric arithmetic.

16 Conclusions, sources, and limitations

Mathematical tests and balances check the implementation. They are not experimental validation, test-bench calibration, or evidence of emissions compliance.

The original assignment, adopted assumptions, numerical results, and external sources remain distinct. HTML, PDF, CSV, and the native document are generated from the same state.

Conclusions concern the applicable checks for the current variant. Installed EPA reference maps are used only within their data domains. Independent testing, complete spatial strength analysis, verified fatigue life, and emissions require separate evidence.

> **Available output-shaft power:** 110 kW. The adopted value is referenced directly from the preceding result.

Relative overall requirement residual.

> **Relative overall requirement residual**
>
> where P_{calc} — Available output-shaft power, W; P_{req} — Required output-shaft power, W
>
> $$\delta \mathrm{P} = \frac{\left(P_{\mathrm{calc}} - P_{\mathrm{req}} \right)}{P_{\mathrm{req}}} \tag{16.1}$$
>
> Substitute the current values in SI units:
>
> $$\delta \mathrm{P} = \frac{\left(110000 - 110000 \right)}{110000} = 2{,}225 \times 10^{-6}$$
>
> The result passes to dependent calculations without being entered again.

Table 16.1. Requirement checks.

Preliminary check	State	Value / limit
Connecting rod: illustrative Goodman criterion	Passed under the assumptions	0,5576 / 1
Connecting rod: Euler only within the elastic domain	Passed under the assumptions	34,55 kN / 64,01 kN
Piston pin: illustrative bending	Passed under the assumptions	239,4 MPa / 600 MPa
Piston pin: mean pressure	Passed under the assumptions	49,35 MPa / 100 MPa

Preliminary check	State	Value / limit
Equivalent shaft: torsion only	Passed under the assumptions	53,83 MPa / 150 MPa

Specified at the output shaft:
110 kW

Calculated with limits:
110 kW

Calculated D × S:
91,86 × 87,27 mm

Displacement / modelled efficiency:
2,314 L / 35,45%

Table 16.2. Requirements, calculated values, and criterion explanations.

Check and status	Value and criterion	Explanation
Engine brake power after adopting geometry **Passed in the model**	110 kW ≥ 110 kW	Brake power after geometry adoption must meet the engine requirement. Numerical tolerance is 0.005 %; this is not a power margin.
Design maximum-pressure limit **Passed in the model**	6,902 MPa ≤ 14 MPa	The largest pressure in the modelled cycle is compared with the specified design limit.
Design mean piston-speed limit **Passed in the model**	16 m/s ≤ 25 m/s	Mean piston speed at the calculation speed must not exceed the design limit.
Single-cylinder mass-balance residual **Passed in the model**	-5.421×10^{-19} kg; |residual| < 10^{-9} kg	Difference between cylinder mass change and total air, exhaust, and fuel flow. This checks numerical consistency, not physical mass loss.
Engine thermal-balance residual **Passed in the model**	-4.075×10^{-13} kW; |residual| < 0.01 W	Difference between supplied energy and all included engine energy flows. Zero is expected only within numerical precision.
Agreement between torque integral and p–V work **Passed in the model**	$2,303 \times 10^{-8}$ % < 0,1 %	Relative difference between p–V work and the torque-integral work. Checks agreement between two calculation methods.
Heat-release and injected-fuel-mass integrals **Passed in the model**	Heat 0.003142 %; fuel 0 %	Both relative residuals must be < 0.2 %: heat-release-law integral and injected-fuel-mass integral.
Output power at the specified point **Passed in the model**	110 kW ≥ 110 kW	Available system power at specified point I-6 / I-7. Component limits are included; electric assistance uses initial SOC. Numerical tolerance is 0.005 %.

> An engineering model with explicit assumptions, not test-bench validation or Euro certification.
> Four-stroke single-zone gas; prescribed injection, heat-release laws, and reservoirs. Gas properties are effective, without chemical equilibrium.
> Prescribed throttling $p_{in}=p_{in0}(0.18+0.82L)$; fixed heat release.

References

1. Kolchin A. I. Calculation of Automotive and Tractor Engines. 4th edition, p. 104. [Original-language source]
2. Automobile. Vehicle Design Calculation. Initial engine-speed table for categories M1–M3 and N1–N3. 2026.
3. Thermal Calculation of Petrol and Diesel Engines: Educational Examples. [Original-language source]
4. Theory and Design of Automotive Engines. Chapter 2: Two-Stroke and Four-Stroke Cycle SI Engines.
5. UNECE WLTP: Technical Report Phase 1b, Annex 8 (PEV / OVC-HEV); UN GTR 15 Amendment 4. WLTC — cycle; WLTP — test procedure. URL: https://wiki.unece.org/download/attachments/29229449/GRPE-72-02%20-%20Technical%20Report%20Phase%201b%20-%20Final%20version.pdf?api=v2. URL: https://wiki.unece.org/download/attachments/54427738/GTR%2315_ECE-TRANS-WP29-2017-099e_Amd4.pdf?api=v2.
6. United States Environmental Protection Agency. ALPHA map packages: A25A-FKS, N57, L15B7, Geely Miller, Toyota TNGA Hybrid, Chevrolet Bolt. 2018–2024.
7. Editable assumptions for preliminary electric-drive design
8. European Commission, European Alternative Fuels Observatory. EU classification of vehicle types: categories M and N. URL: https://alternative-fuels-observatory.ec.europa.eu/general-information/vehicle-types.
9. Automobile: cycle-energy.js and $\mathrm{vehicle}_{\mathrm{prototypes}}$.csv; electric-energy-balance adaptation
10. Supplied BEV procedure CSV files and model assumptions
11. Editable BEV loss-of-speed model
12. Roylance D. Fatigue. Cambridge : Massachusetts Institute of Technology, 1999. Kirtley J. L. Electric Machines. Course Notes 7, 9. Cambridge : Massachusetts Institute of Technology, 2013.
13. Opanasiuk Ye. H. Course Project Guidelines for Automotive Engines, specialism 7.090258 Automobiles and Vehicle Operation. Zhytomyr: ZhITI, 2003. 100 pp. [Original-language source]
14. Internal Combustion Engine Handbook. P. 452–455.
15. Slynko, Sukhonos, Slynko. Thermal and Dynamic Calculation of Internal Combustion Engines, 2024 [Original-language source] URL: https://eir.zp.edu.ua/items/707c3e64-76ed-4416-be64-df540ee931d7.
16. United States Environmental Protection Agency. ALPHA maps: Combining Data into Complete Engine ALPHA Maps; Combining Data into Complete eMotor ALPHA Maps. URL: https://www.epa.gov/vehicle-and-fuel-emissions-testing/combining-data-complete-engine-alpha-maps. URL: https://www.epa.gov/vehicle-and-fuel-emissions-testing/combining-data-complete-emotor-alpha-maps.
17. Toyota. New Toyota Auris. 2015. URL: https://media.toyota.co.uk/new-toyota-auris/.
18. Heywood J. B. Internal Combustion Engine Fundamentals. New York : McGraw-Hill, 1988. P. 33.

19. Regulation (EU) 2018/858 of the European Parliament and of the Council of 30 May 2018. Annex I: General definitions, criteria for vehicle categorisation, types of vehicle and types of bodywork. URL: https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX%3A32018R0858.

20. U.S. Department of Energy, Alternative Fuels Data Center. Hybrid Electric Vehicles: parallel and series configurations, battery charging. URL: https://afdc.energy.gov/vehicles/electric-basics-hev.

21. Cantera documentation. Reactors and Reactor Networks: homogeneous thermodynamic control volumes. URL: https://cantera.org/stable/reference/reactors/index.html.

22. NPTEL. Architecture of Hybrid and Electric Vehicles, Module 3, Lecture 5: energy conversion and engine losses. URL: https://archive.nptel.ac.in/content/storage2/courses/108103009/module3/lec5/2.html.

23. Brooker et al. FASTSim: A Model to Estimate Vehicle Efficiency, Cost and Performance (2015) URL: https://www.nrel.gov/docs/fy15osti/63623.pdf.

24. Webasto Standard Battery Pro 40: technical data URL: https://www.webasto.com/en-int/battery/standardized-battery-and-thermo-management/standard-battery-pro-40.html.

25. EMRAX 228: separate continuous and peak ratings URL: https://emrax.com/e-motors/emrax-228/.

26. WLTC Class 3b: JRCSTU/wltp; normalised copy Automobile URL: https://github.com/JRCSTU/wltp.

27. BMW: 3 Series technical description URL: https://www.press.bmwgroup.com/usa/article/detail/T0216443EN_US/the-new-bmw-3-series-sedan-and-sports-wagon?language=en_US.

28. Brogen: electric motors for light commercial vehicles URL: https://brogenevsolution.com/electric-motors-for-lcvs/.

29. Porsche Taycan glossary: 800 volt system URL: https://newsroom.porsche.com/en_US/products/taycan/glossary-18550.html.

30. UNECE GRPE/2022/12: PEV sequence, terminal-energy weighting and range URL: https://unece.org/sites/default/files/2022-03/ECE-TRANS-WP.29-GRPE-2022-12e.pdf.

31. UNECE EVE-62-07: WLTP constant-speed discharge segments URL: https://wiki.unece.org/download/attachments/198673568/EVE-62-07e%20-%20HDV%20GTR%20Japanese%20UBE%20measurement.pdf?api=v2.

32. EC/JRC: WLTC Class 3a and 3b GTR speed data URL: https://github.com/JRCSTU/wltp/tree/master/wltp/cycles.

33. UNECE: speed capping with phase distance compensation URL: https://unece.org/DAM/trans/doc/2017/wp29grpe/ECE-TRANS-WP29-GRPE-2018-02e_track.pdf.

34. UNECE: PEV speed-trace tolerance and 4-second break-off URL: https://wiki.unece.org/download/attachments/56591396/WLTP-SG-EV-18-07%20Drive%20Trace%20Index%20EV%20JP.pdf?api=v2.

35. BMW i4 xDrive40: technical specifications, high-voltage battery 398.5 V. URL: https://www.press.bmwgroup.com/global/article/attachment/T0442767EN/617124.

36. Hyundai IONIQ 5: 800 V battery system and 400/800 V charging compatibility. URL: https://www.hyundai.com/eu/en/models/ioniq5/performance.html.